

Rotational Dynamics – torques - Levers - Center of mass -

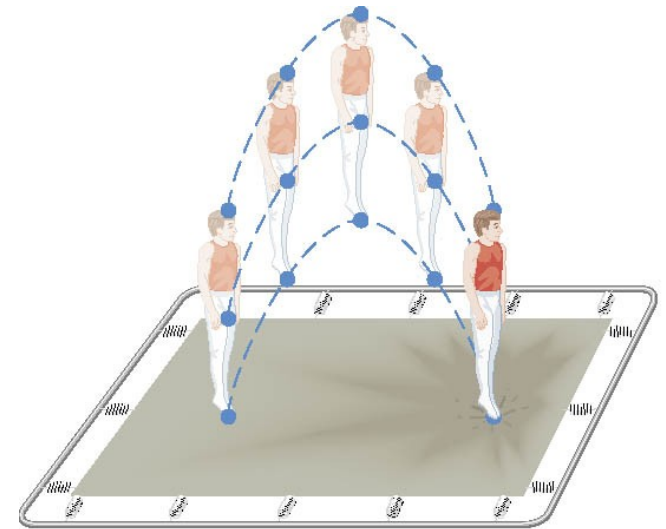


What causes an object to have an *angular acceleration*
Torque is a measure of the force that can cause an object to rotate about an axis. Just as force is what causes an object to accelerate in linear kinematics

Source: Physics / Edition 8
by John D. Cutnell, Kenneth W. Johnson, Cutnell

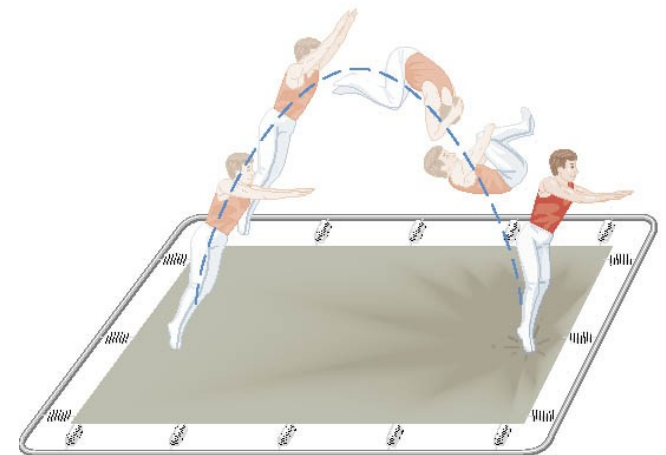
-The physics of everyday phenomena: a conceptual introduction to physics
by W. Thomas Griffith, McGraw-Hill

In pure translational motion, all points on an object travel on parallel paths.



(a) Translation

The most general motion is a combination of translation and rotation.



(b) Combined translation and rotation

According to Newton's second law, a net force causes an object to have a linear acceleration.

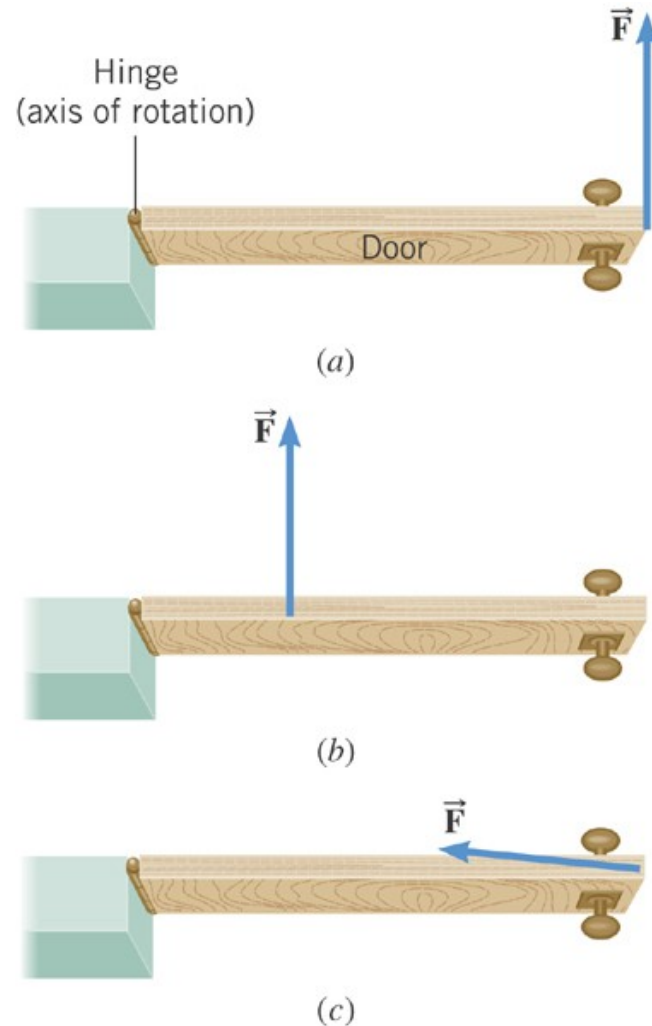
What causes an object to have an *angular acceleration*?

TORQUE τ

MEASURES HOW EFFICIENT A FORCE IS IN APPLYING A TWIST :

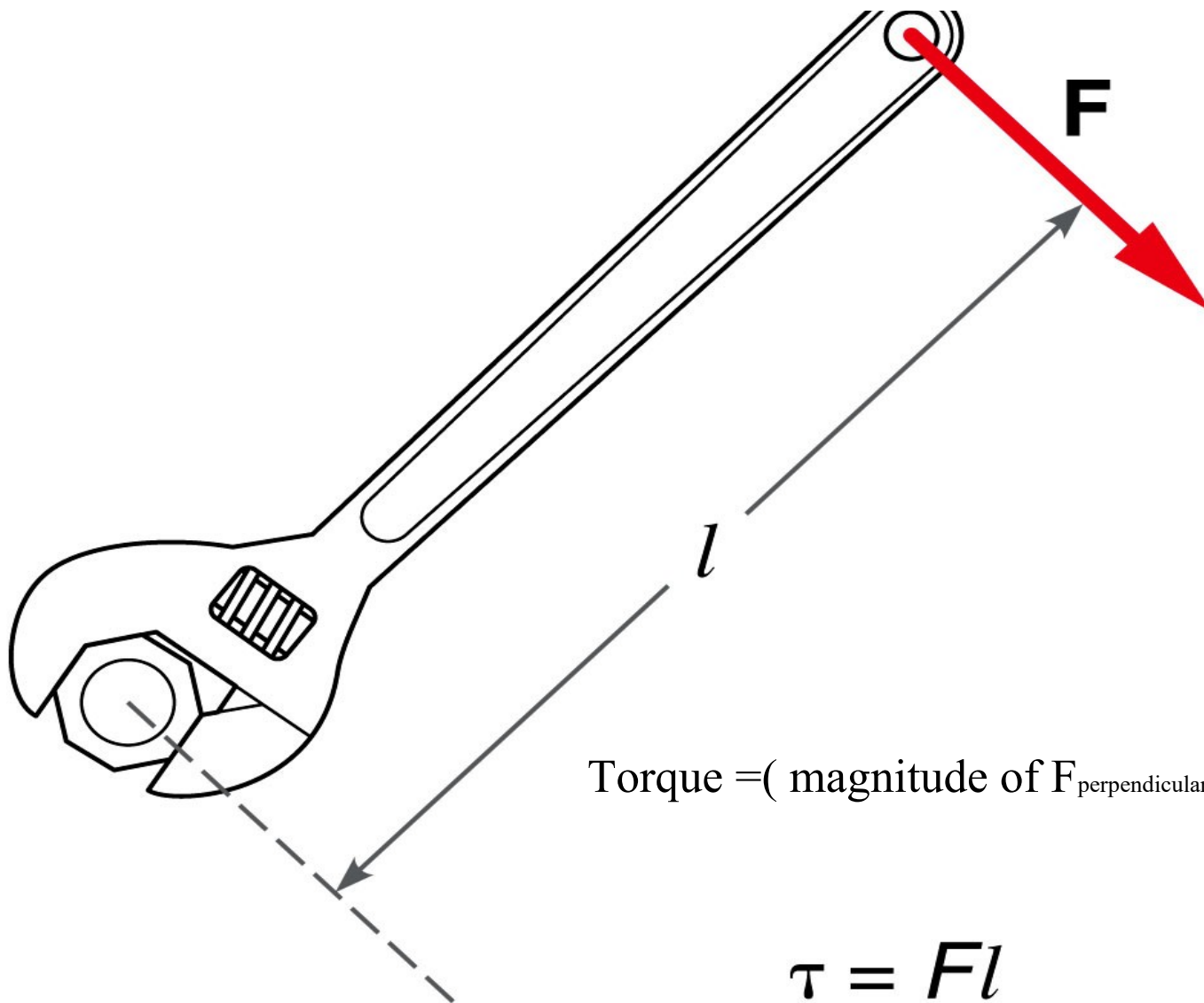
**Depend on the distant to the axis of rotation
and the direction of the force.**

Torque = how efficient is a force in rotating a rigid object (twist)
In which case the torque is the largest ? (more efficient in opening the door).



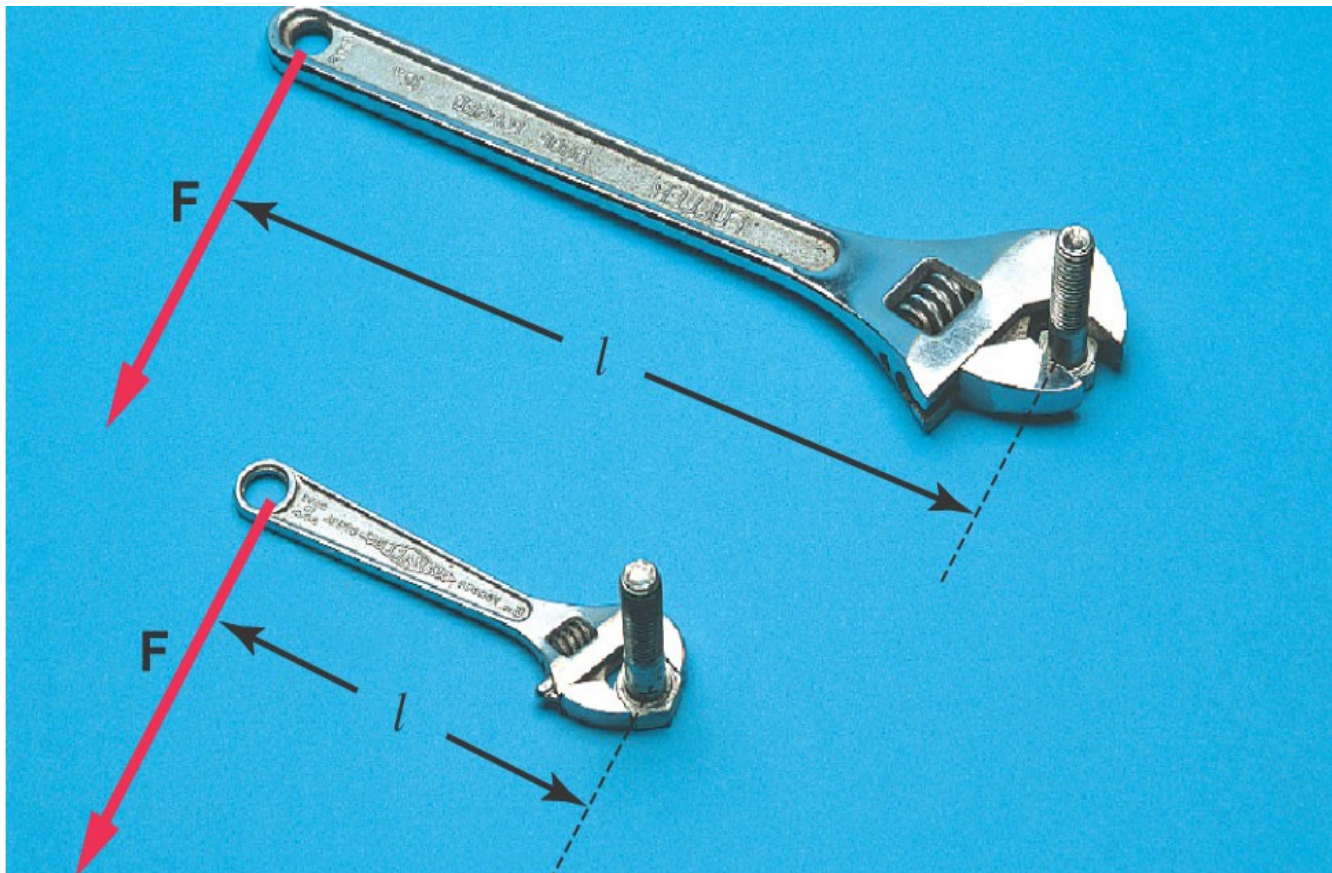
The amount of torque depends on where and in what direction the force is applied, as well as the location of the axis of rotation.

Any Suggestion for the expression of a torque when the force is normal
To the door ?



To be more efficient you can: increase F or increase F a better twist.

If $F=50\text{N}$ and the length of the wench is 24cm , find the torque. (cm to m)



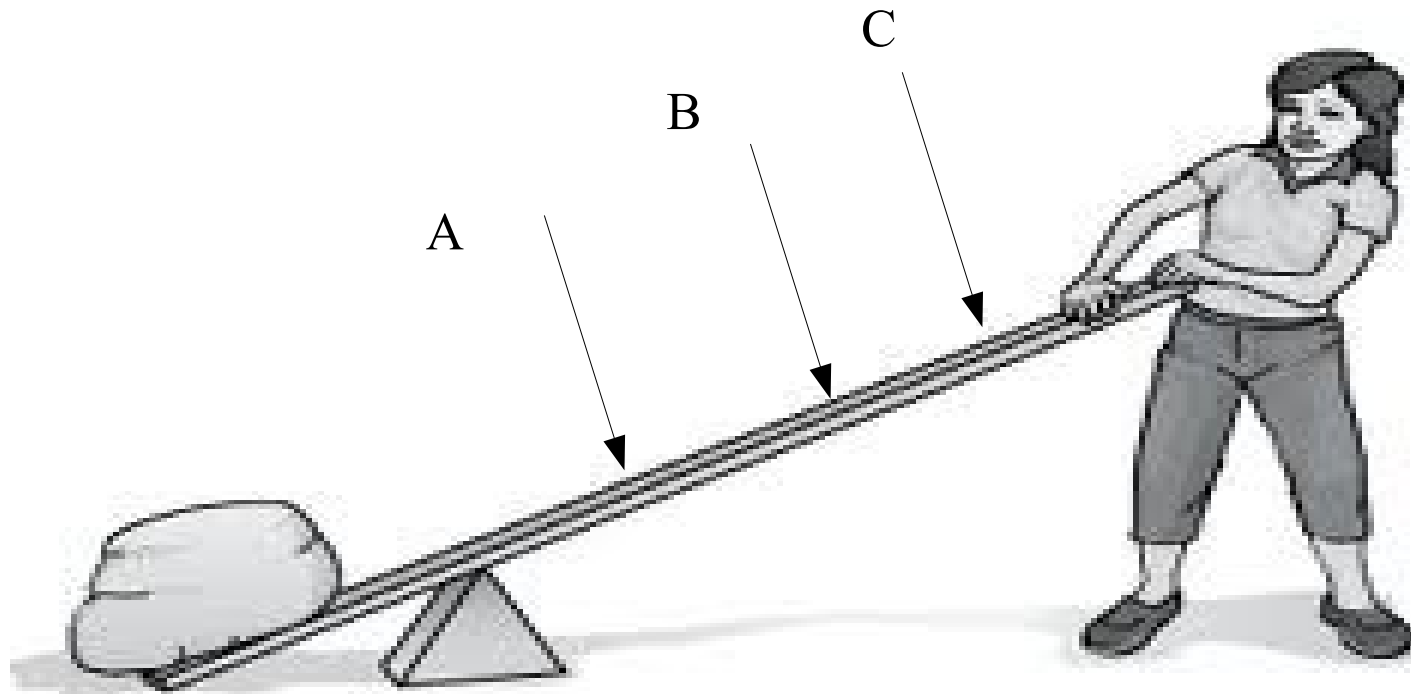
If l is divided by 2, the new torque :

- A) is the same
- B) is double
- C) is half

So why do we use long wrench ?

Where to push to make the torque the largest ?
(will make easier to push) A or B or C ?

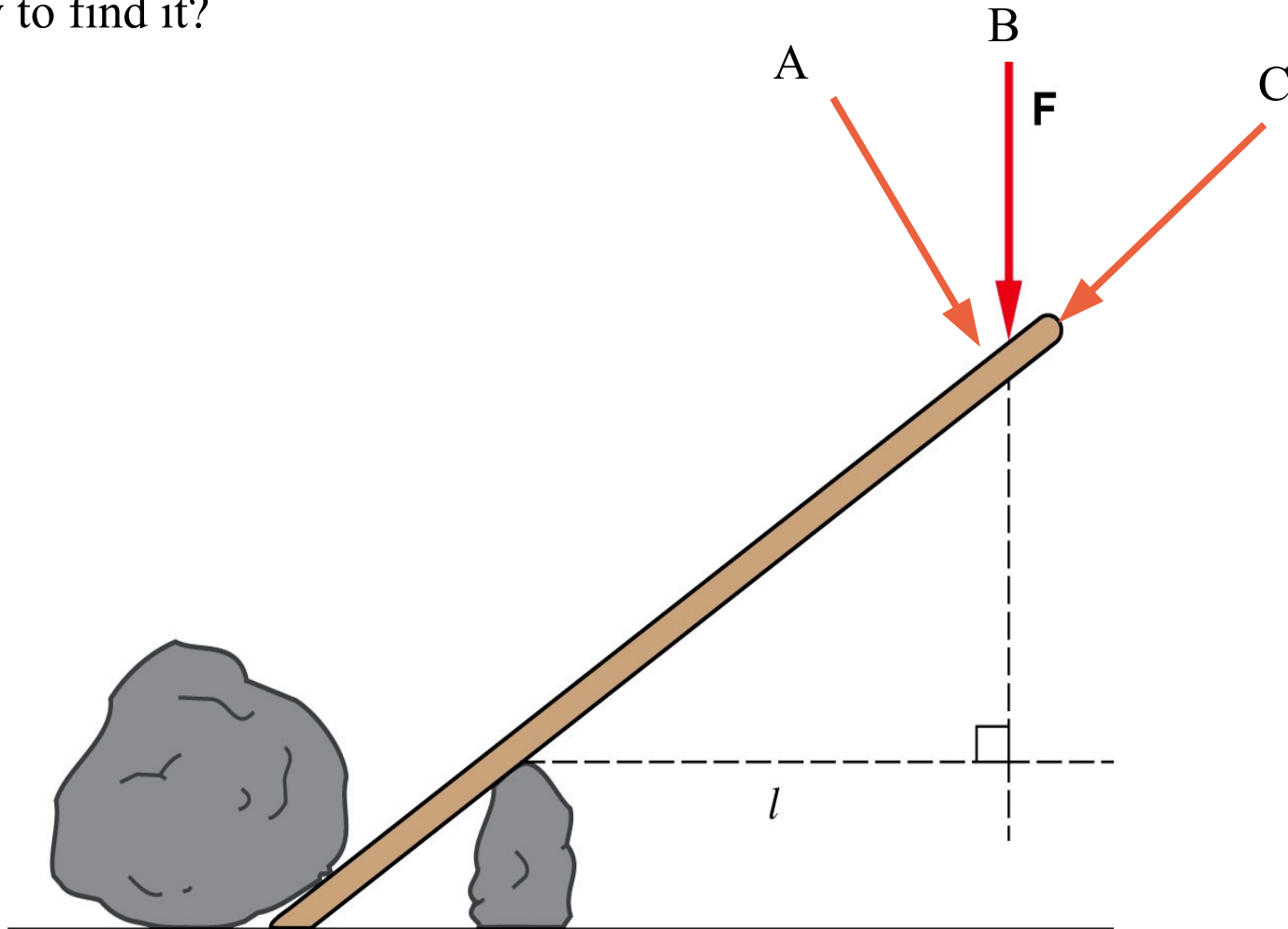
The force applied is the same.



demo

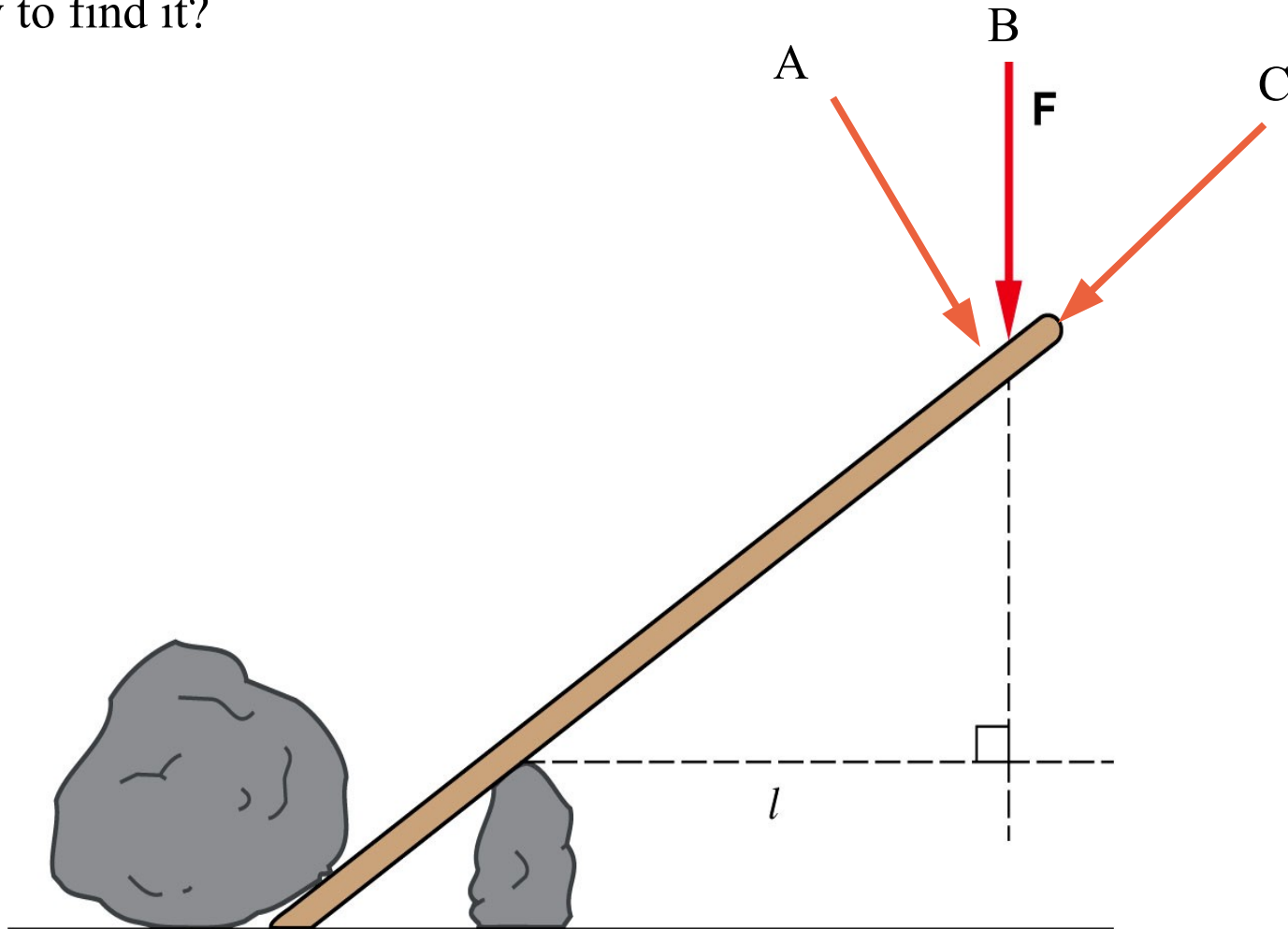
Which direction will produce the larger twist ? A? B? C?
(more efficient for The same magnitude?)

What is the the level arm l ?
How to find it?



Which direction will produce the larger twist ? A? B? C?
(more efficient for The same magnitude?)

What is the the level arm l ?
How to find it?



Torque - definition

Magnitudes:

$$\text{Torque} = \tau = \mathbf{r} \times \mathbf{F}_{\perp}$$

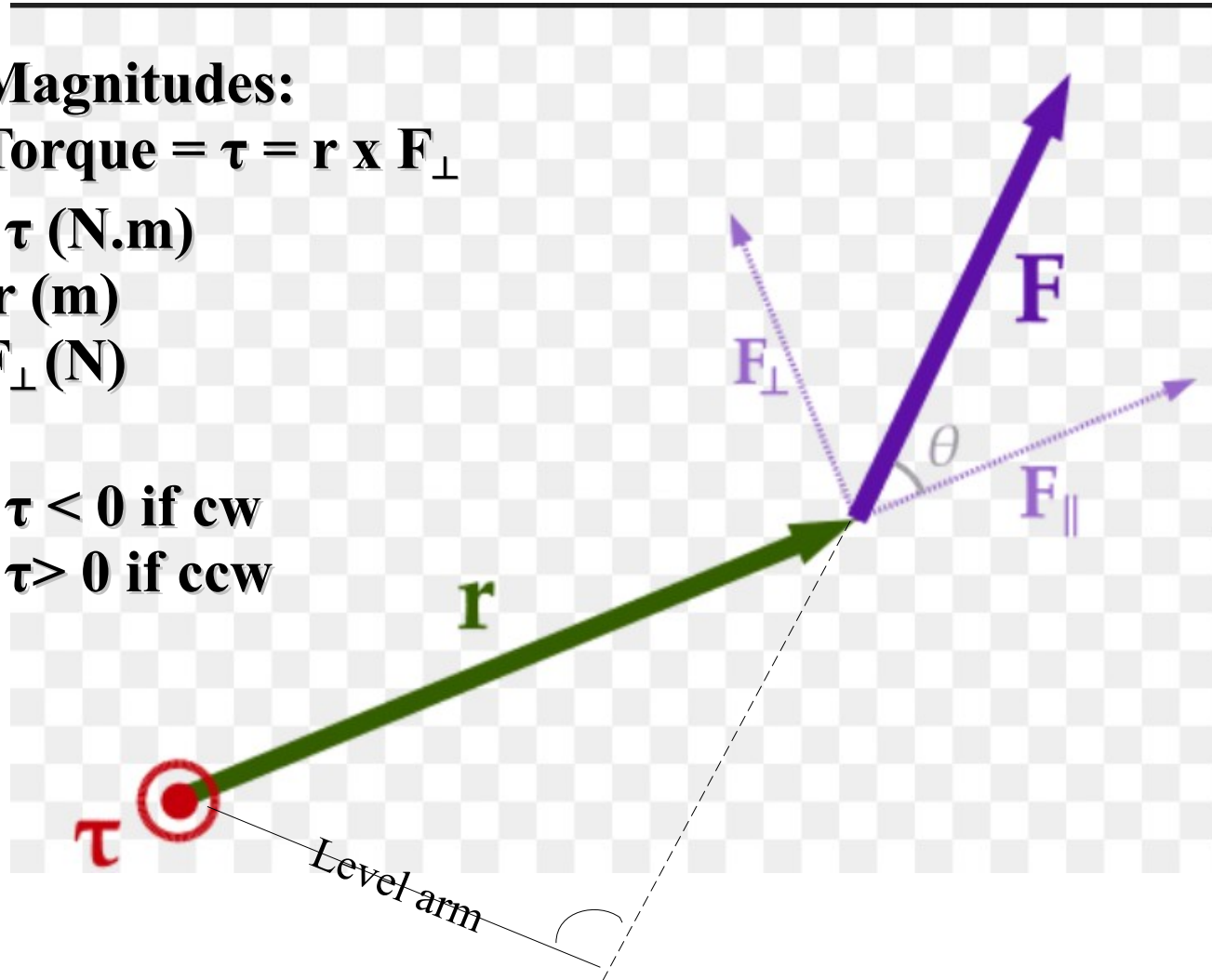
. τ (N.m)

r (m)

$F_{\perp}$ (N)

. $\tau < 0$ if cw

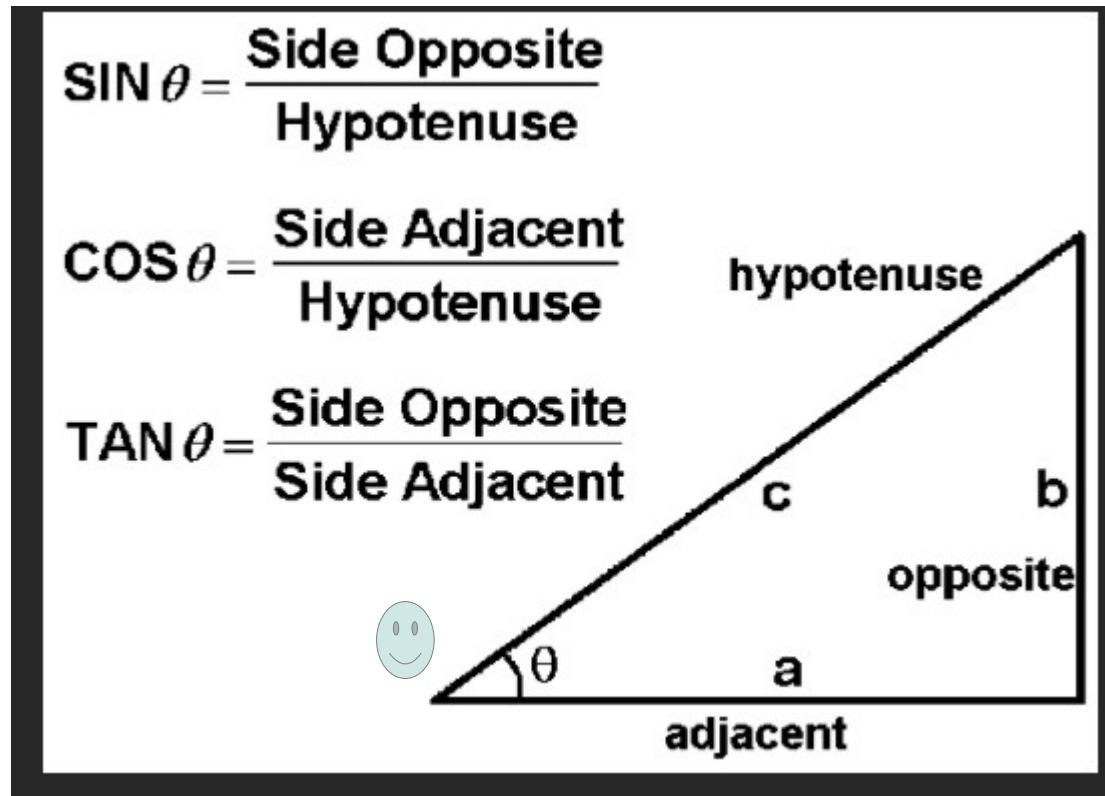
. $\tau > 0$ if ccw



Show the level arm. Either torque = $(F \sin(\theta)) \times r$ or
Torque = $F \times \text{level arm} = F \times (r \sin(\theta))$

To find adjacent (close leg to ☺): use **co**☺ne(..) and multiply by hypotenuse (long leg c)

To find the opposite (opposite leg to ☺) : use **s**☺e (..) and multiply by the hypotenuse (long leg c)



http://www.funmaths.com/worksheets/downloads/view.htm?ws0037_1.gif

http://www.funmaths.com/worksheets/downloads/view.htm?ws0038_1.gif HW for WEdnesday₁₁

Suppose the angle is 40°

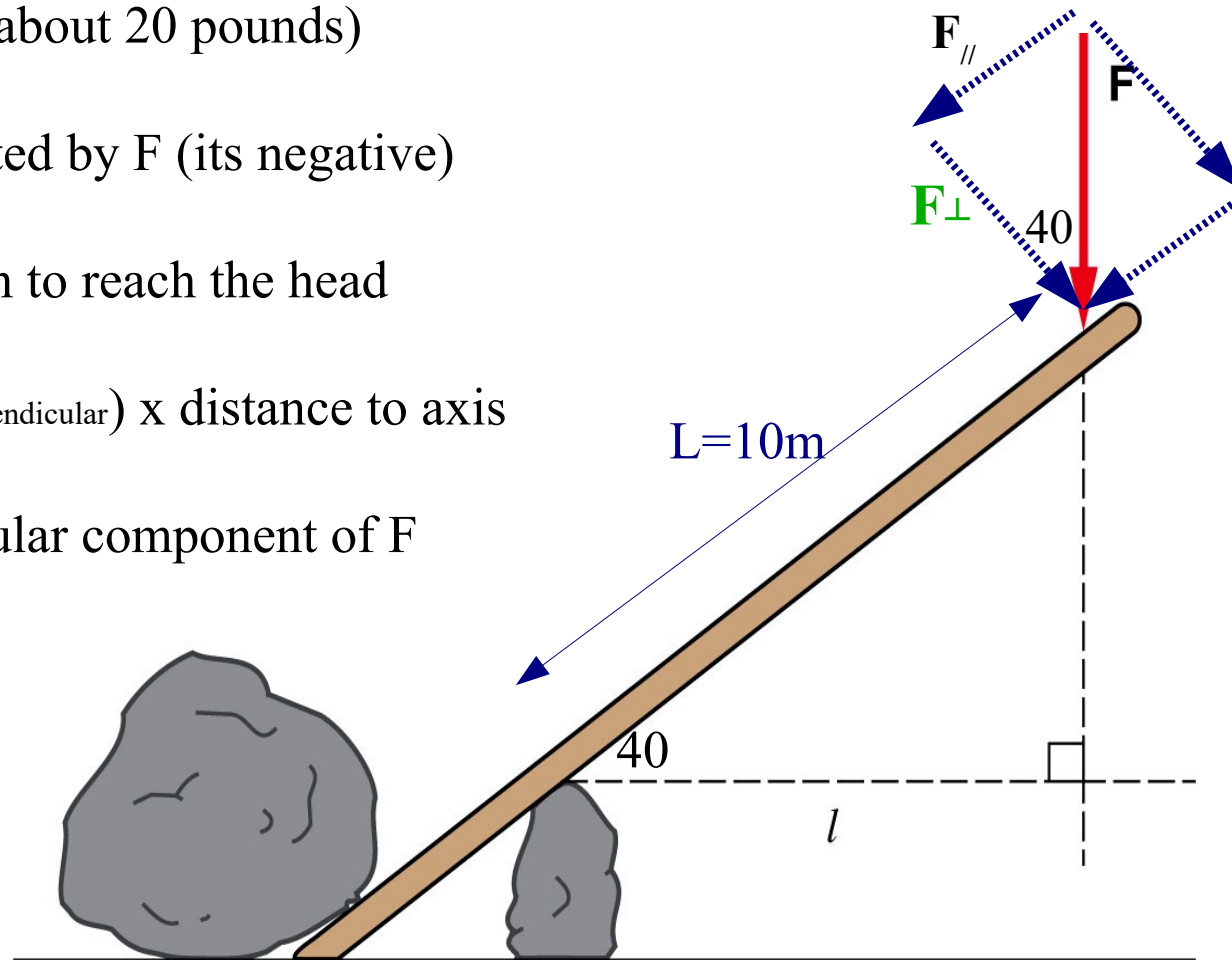
Suppose F is 100N (about 20 pounds)

Find the torque exerted by F (its negative)

From tail fall and run to reach the head

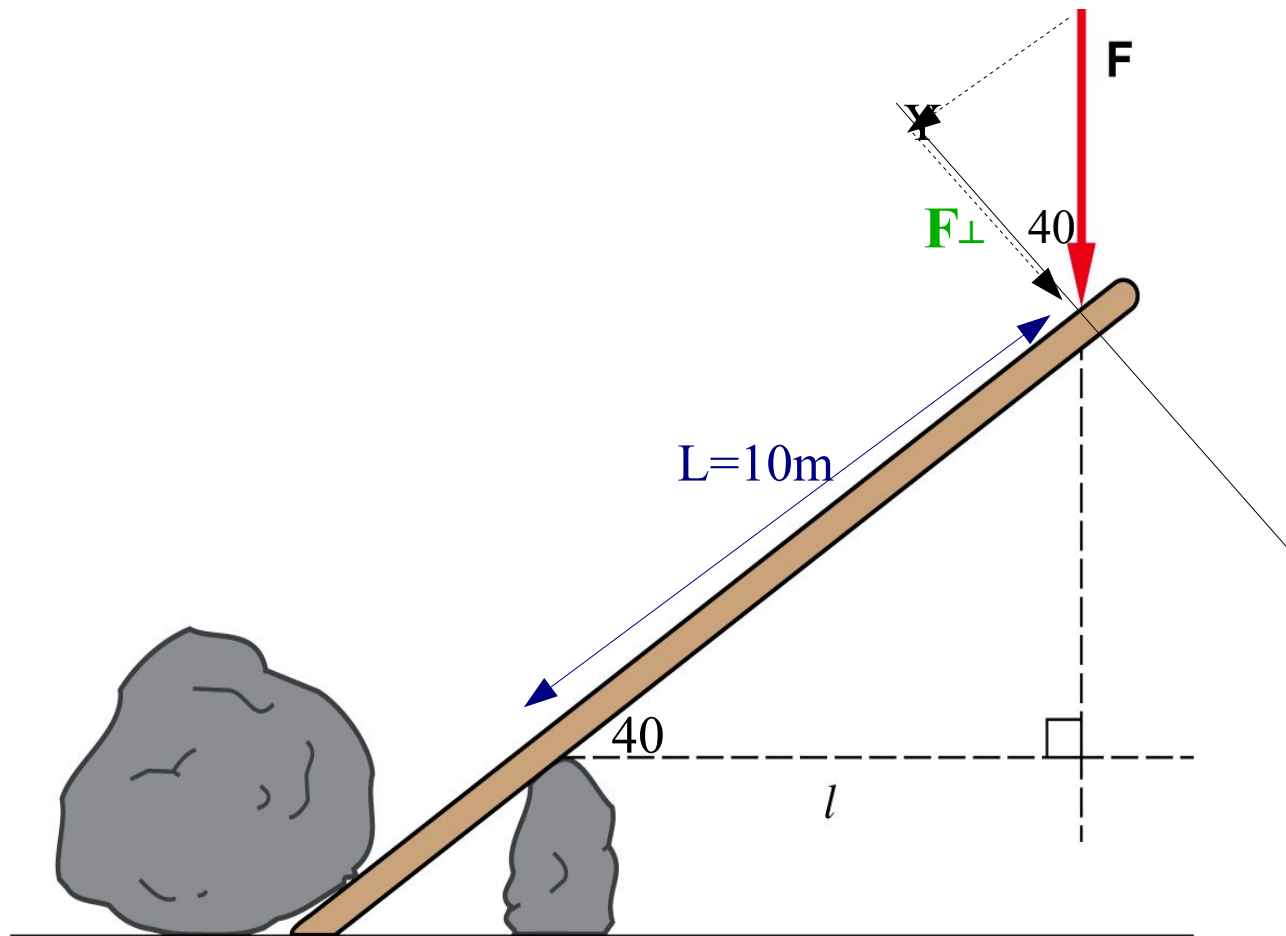
- (magnitude of $F_{\text{perpendicular}}$) x distance to axis

- only the perpendicular component of F is doing the “job”

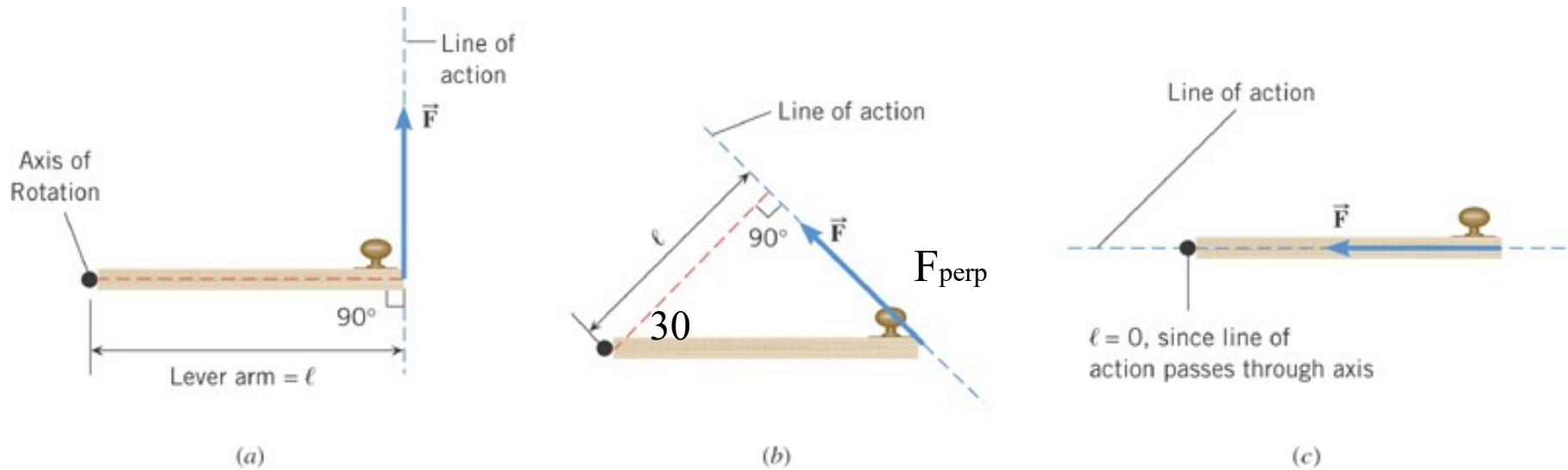


Only the component $F_{\perp}$ (normal) is doing the job.

Or you can run and fall.
Same result.



How to find ℓ the lever arm = perpendicular distance between the axis and the line of action of the force.



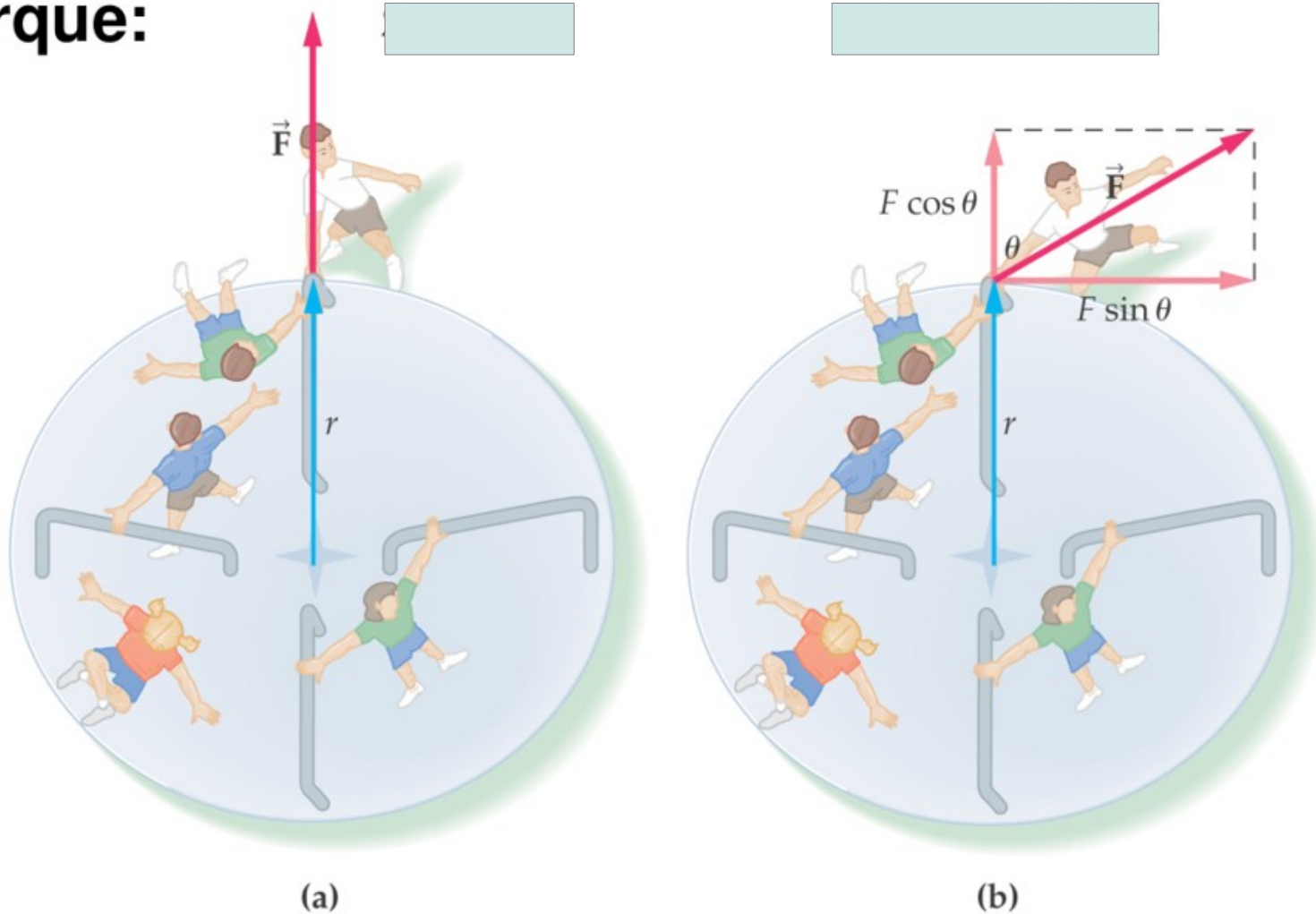
DEFINITION OF TORQUE

$$\tau = F_{\text{perpendicular component}} L$$

Direction: The torque is positive when the force tends to produce a counterclockwise rotation about the axis.

SI Unit of Torque: newton x meter (N·m)

Only the tangential component of force causes a torque:

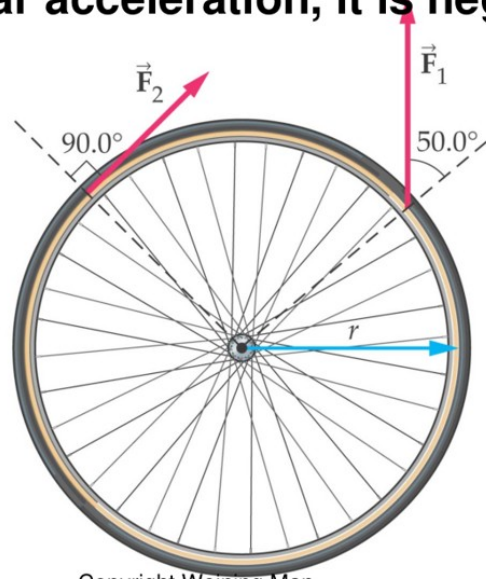


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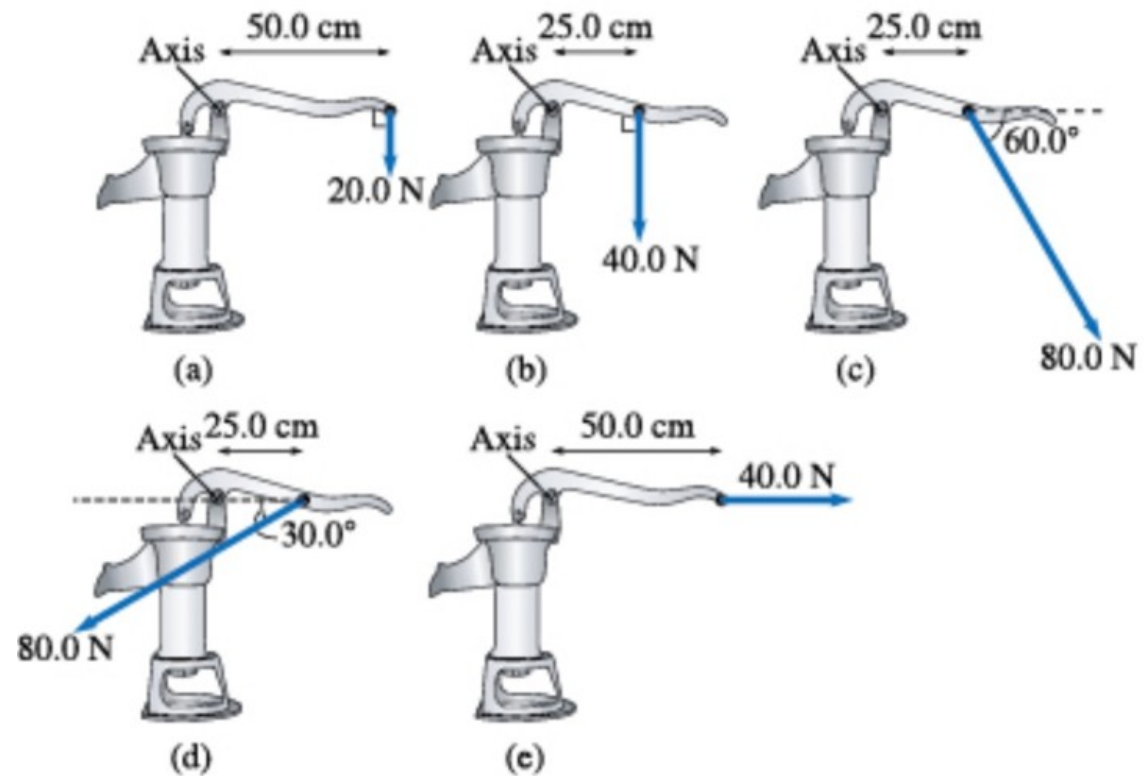
In (a) what is the torque ?

In (b) is the torque: $F r$, $F r \sin(\theta)$, $F r \cos(\theta)$?

angular acceleration, it is negative



Torque of F_1 . Is it sine? Cosine?



9.1.6. An object with a triangular cross-section is free to rotate about the axis represented by the black dot shown. Four forces with identical magnitudes are exerted on the object. Which one of the forces, if any, exerts the largest torque on the object?

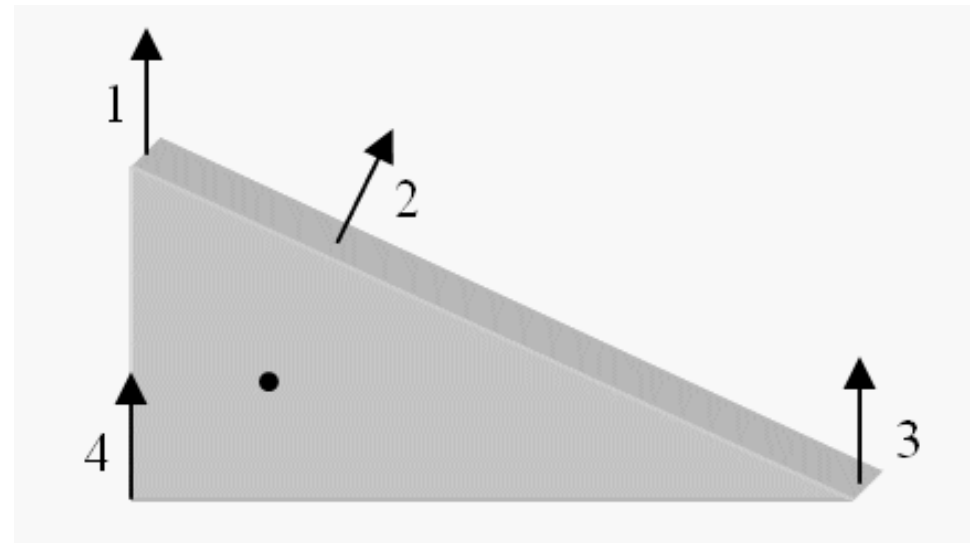
a) 1

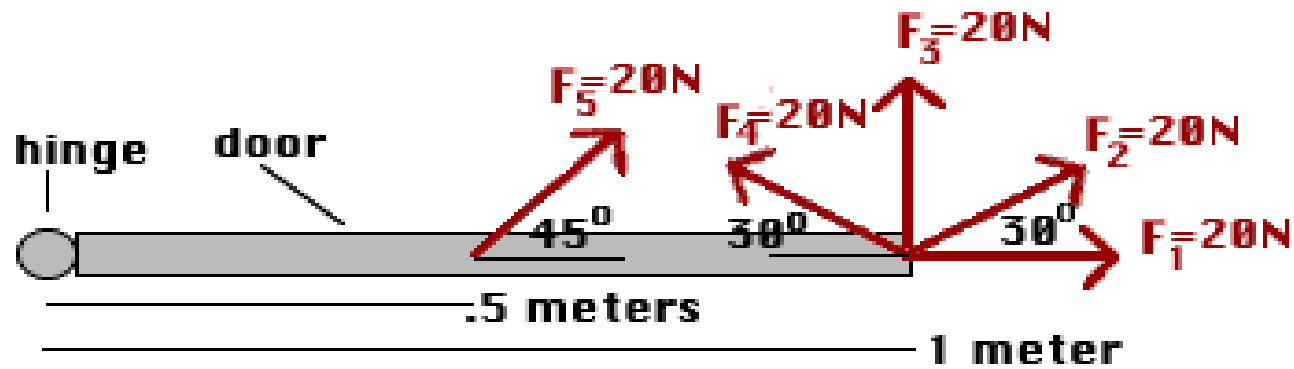
b) 2

c) 3

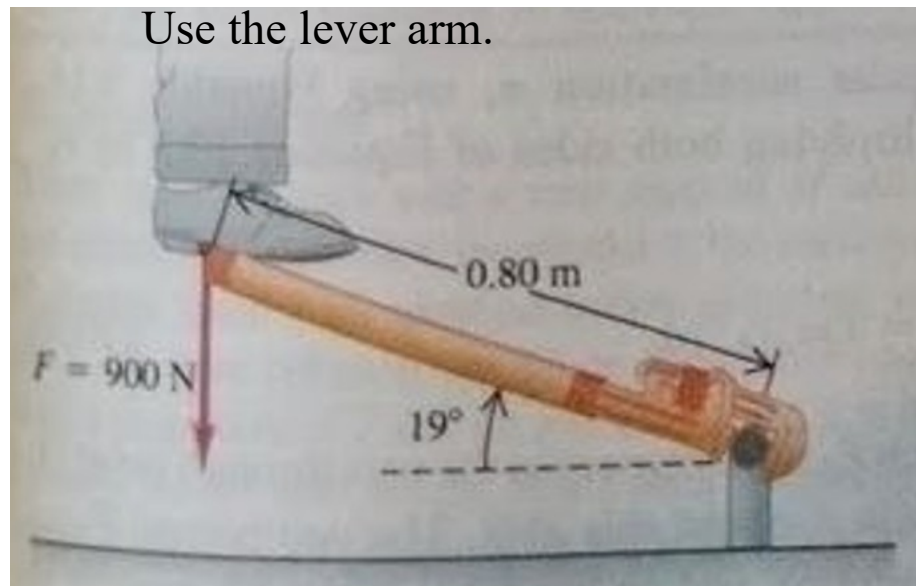
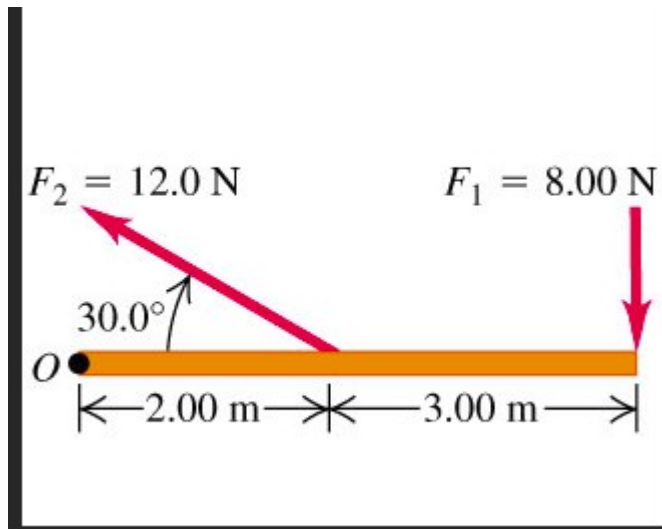
d) 4

e) The same torque is exerted by each force.





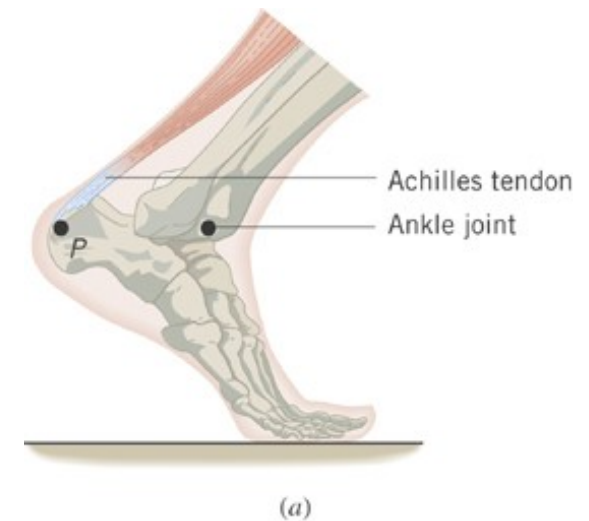
Find the torque exerted by each force.



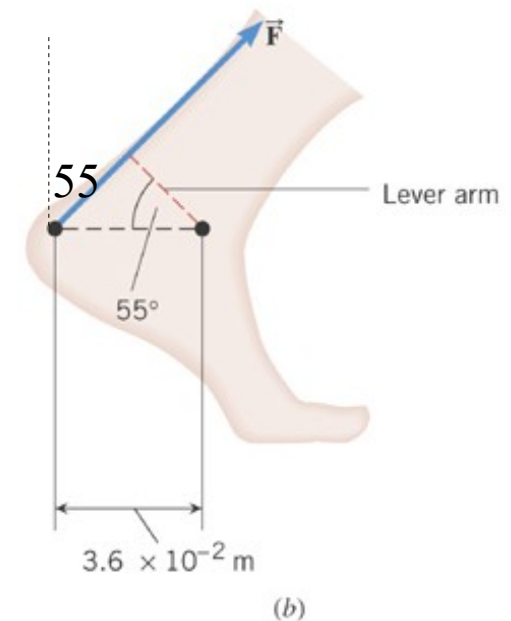
Textbook 10.1

Example 2 The Achilles Tendon

The tendon exerts a force of magnitude 790 N. Determine the torque (magnitude and direction) of this force about the ankle joint.



$$\tau = F_{\text{perp}} \ell \quad \text{or}$$

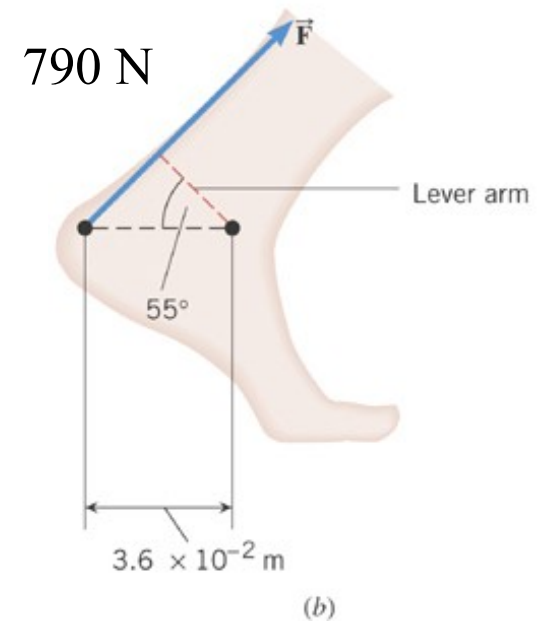
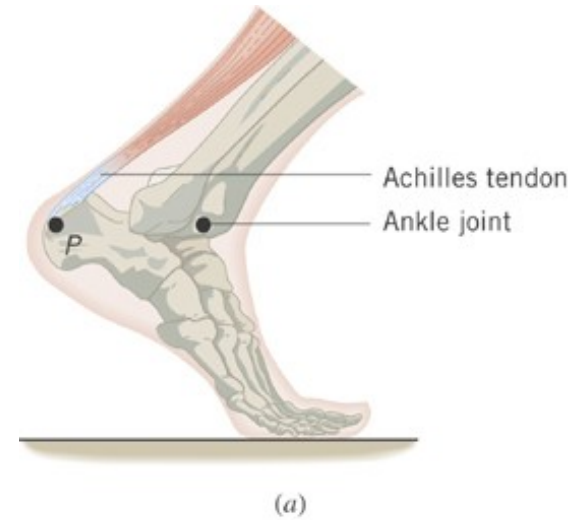


9.1 The Action of Forces and Torques on Rigid Objects

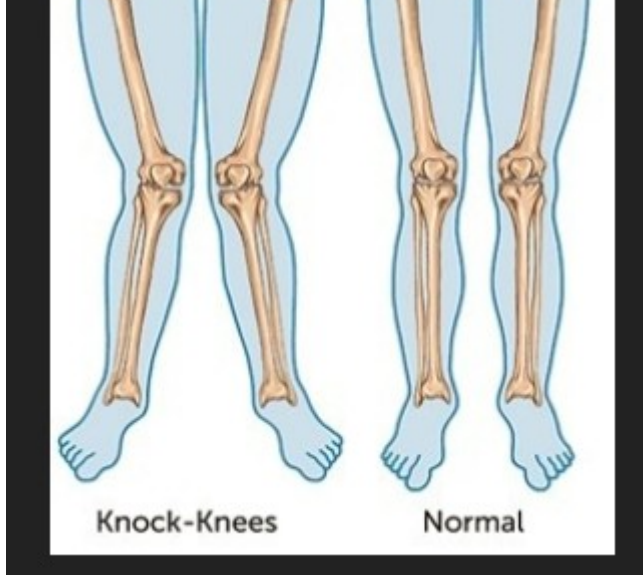
$$\tau = F\ell$$


$$\cos 55^\circ = \frac{\ell}{3.6 \times 10^{-2} \text{ m}}$$

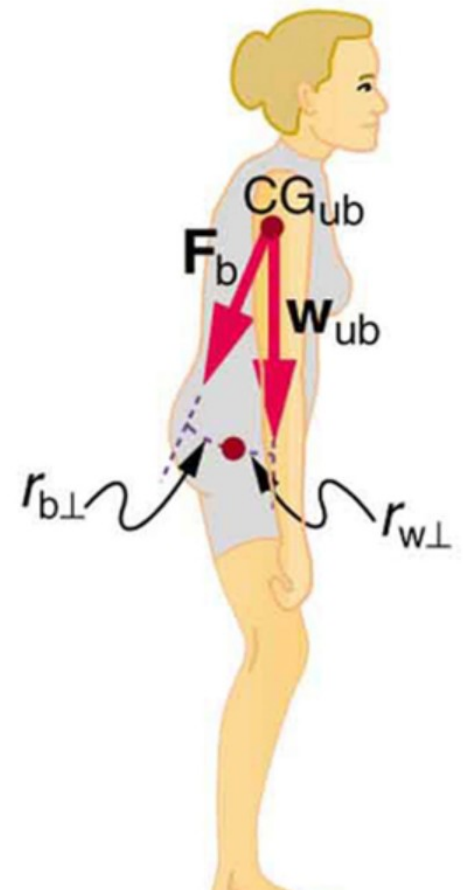
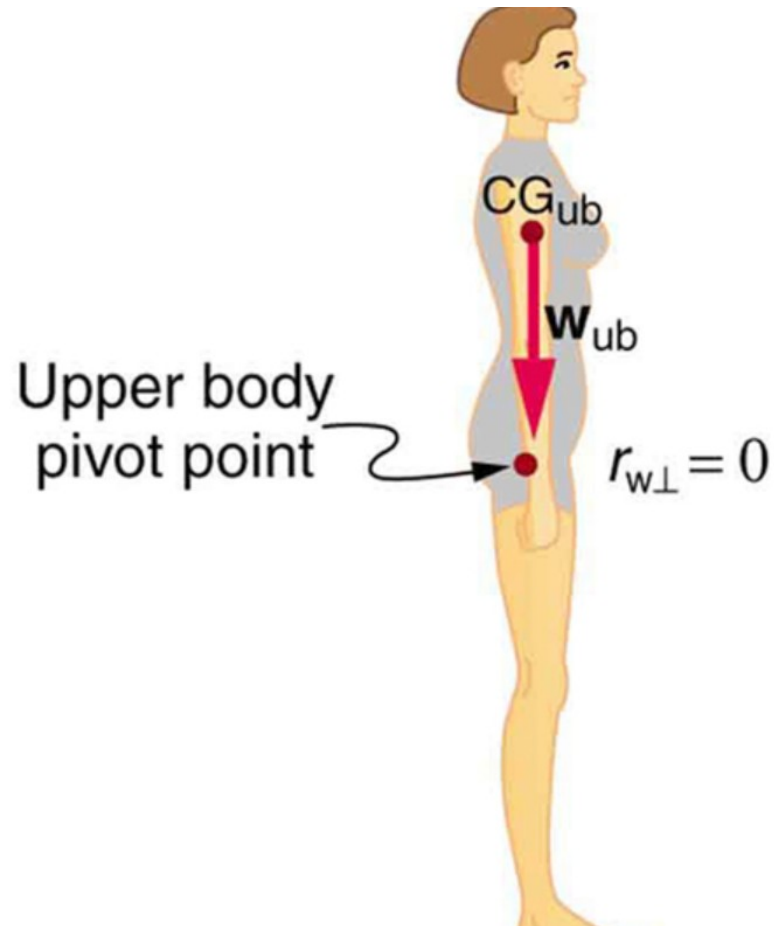
$$\begin{aligned}\tau &= (720 \text{ N})(3.6 \times 10^{-2} \text{ m})\cos 55^\circ \\ &= 15 \text{ N} \cdot \text{m}\end{aligned}$$



Video my student. torque



For the same first, is it better a long or short arm ?



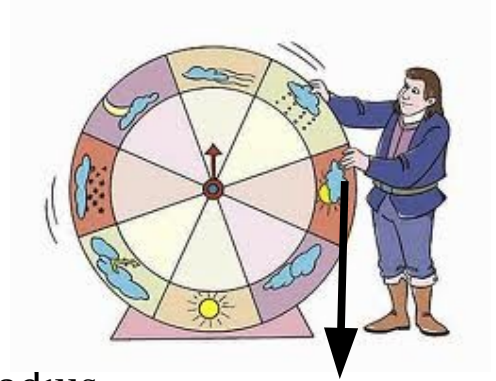
1) The radius of the wheel of fortune is 1.2m, and the operator applies a force of 45N Tangentially to get it spinning. What torque has he supplied ?

2) a 32 kg child sits on a seesaw. If she is 2.2 m from the pivot, what is the torque That her weight exerts, making the seesaw rotate around the pivot ?

3) A torque of 30N.m is required to turn the steering wheel of a car. If the radius Of the wheel is 26cm/ How much force is needed.

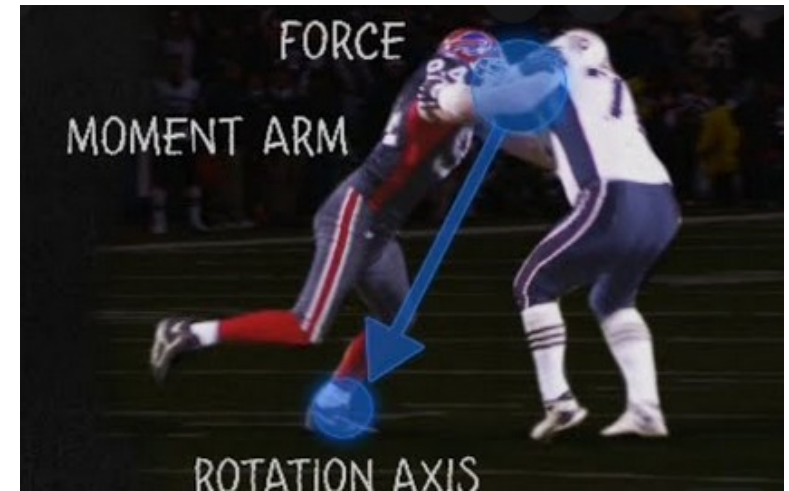
4) You are using a crowbar 2.4 m long, pivoted at 0.20m from the end, to lift at the end A rock that weighs 750N. How much force do you have to exert ?

10) A force of 50N is applied at the end of a wrench handle that is 24cm
The force is applied in a direction perpendicular
To the handle . A) what is the torque applied to the nut by the wrench
B) what would be the torque if the force were applied half
Way up instead of at the end ?



Go over this

LOW MAN WINS



Show video of my student (gym)

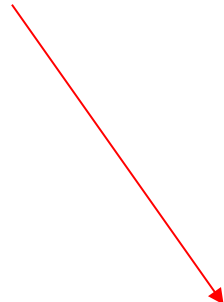
9.2 *Rigid Objects in Equilibrium*

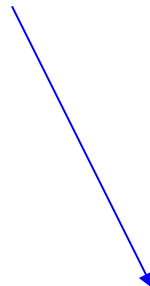
If a rigid body is in equilibrium, neither its linear motion nor its rotational motion changes.

$$a_x = a_y = 0$$

$$\alpha = 0$$


$$\sum F_x = 0$$


$$\sum F_y = 0$$


$$\sum \tau = 0$$

EQUILIBRIUM OF A RIGID BODY

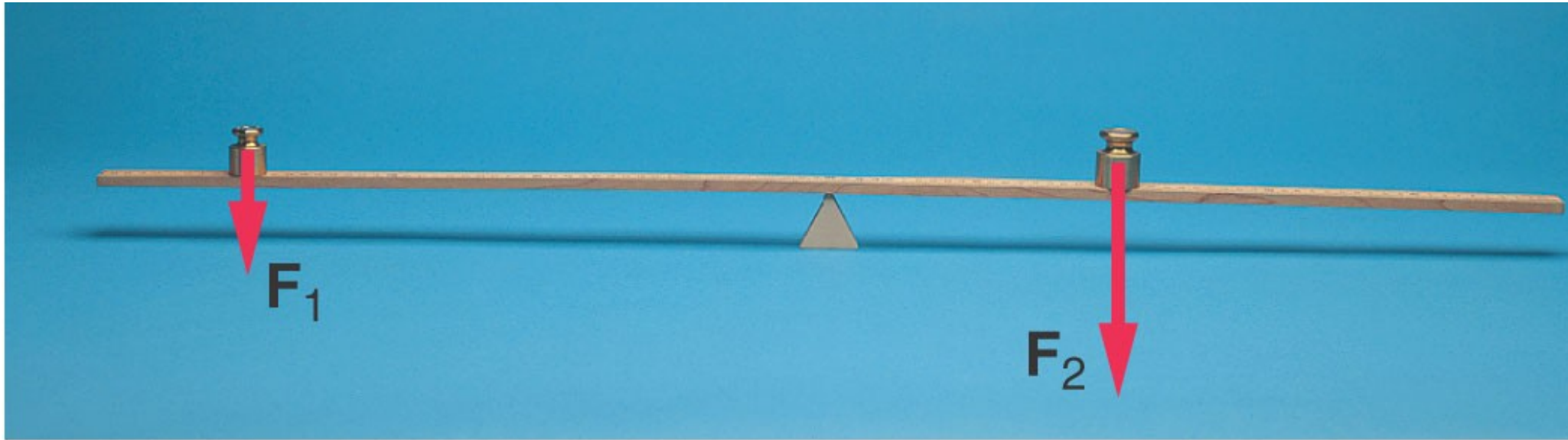
A rigid body is in equilibrium if it has zero translational acceleration and zero angular acceleration. In equilibrium, the sum of the externally applied forces is zero, and the sum of the externally applied torques is zero.

$$\sum F_x = 0$$

$$\sum F_y = 0$$

$$\sum \tau = 0$$

Example of equilibrium:



https://phet.colorado.edu/sims/html/balancing-act/latest/balancing-act_en.html

If mass 2 = 1kg then $F_2 = \underline{\hspace{2cm}} \text{ N}$

If mass 1 = 500g, the $F_1 = \underline{\hspace{2cm}} \text{ N}$

If the masses are in equilibrium :

$$+ F_1 l_1 - F_2 l_2 = 0$$

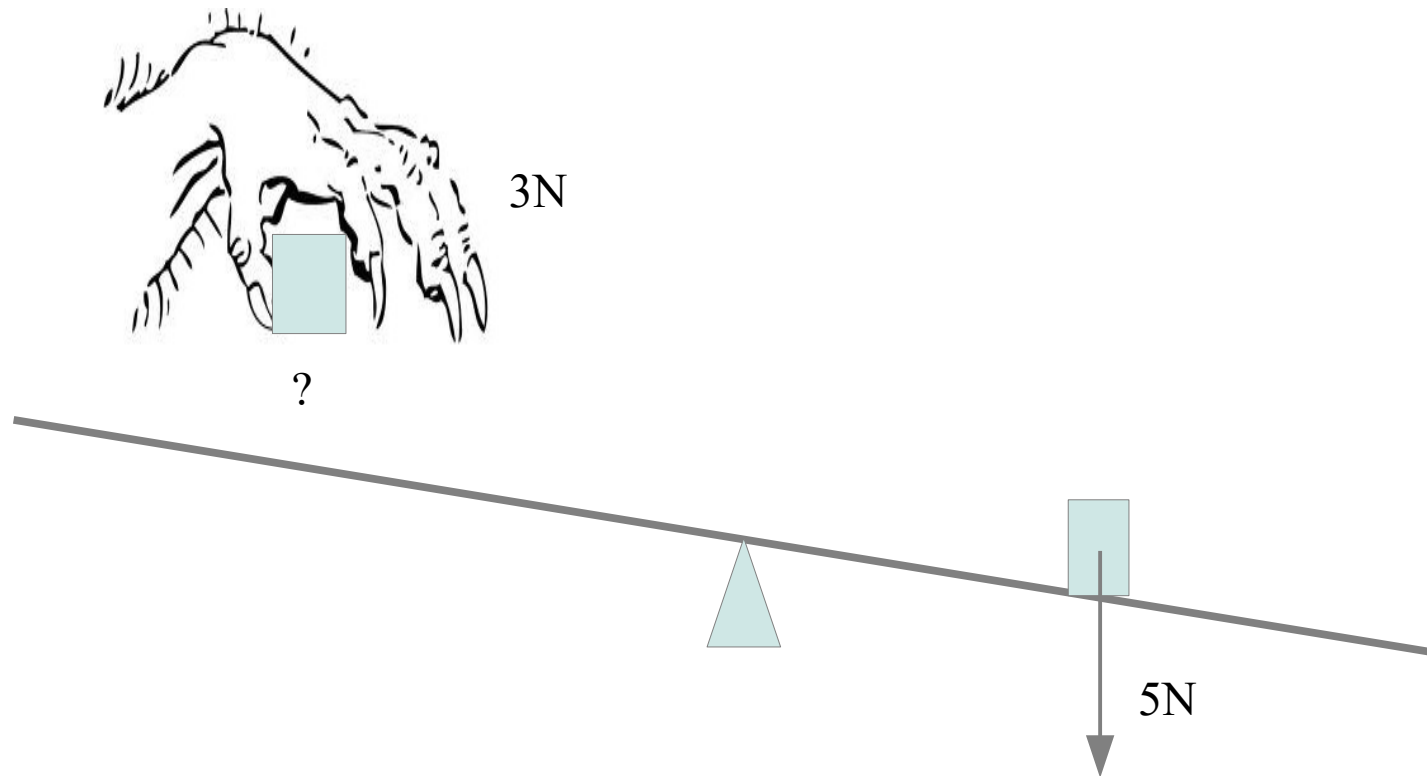
If $d_2 = 10 \text{ cm}$, find d_1 .

Draw a free body diagram and use up = right to find the force exerted by the Fulcrum on the ruler.

Suppose we have a 3N weight that we want to balance against a 5 N weight
On a stick, which is balanced when no mass are in place. The 5N weight is placed
20cm to the right of the fulcrum. (see below)

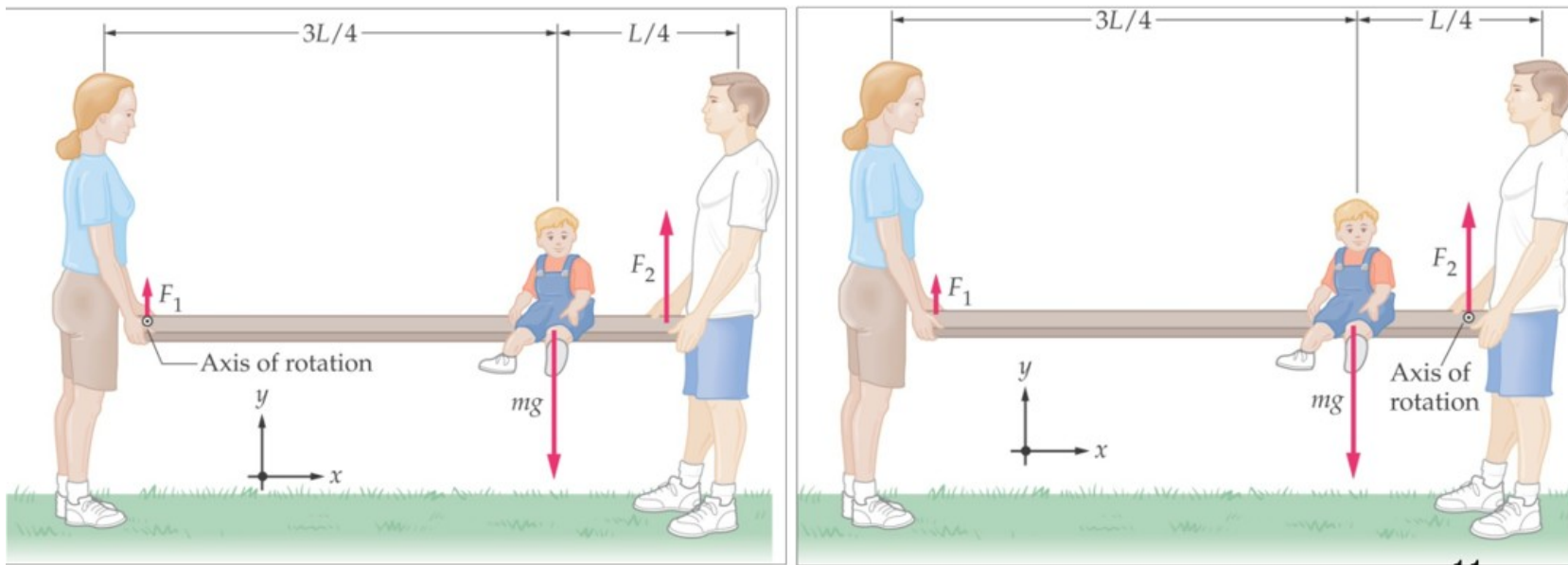
A) What is the torque produced by the 5N weight (cm to m and sign !!)

B)How far would we have to place the 3N weight from the fulcrum to balance the
System ?



11-3 Zero Torque and Static Equilibrium

If the net torque is zero, it doesn't matter which axis we consider rotation to be around; we are free to choose the one that makes our calculations easiest.



Also. The unit does not matter for distance or force. As long as all the forces have the Same units. And all the distances have the same unit.

REMEMBER: EQUILIBRUM MEANS :

I. $\sum(\text{torques}) = 0$

$$\pm F_1 l_1 \pm F_2 l_2 \pm F_3 l_3 + \dots = 0$$

With $F_1, F_2, F_3 \dots$ are the magnitude of the y-components of the forces
(perpendicular components)

l_1, l_2, l_3 are the distances from the fulcrum (axis).

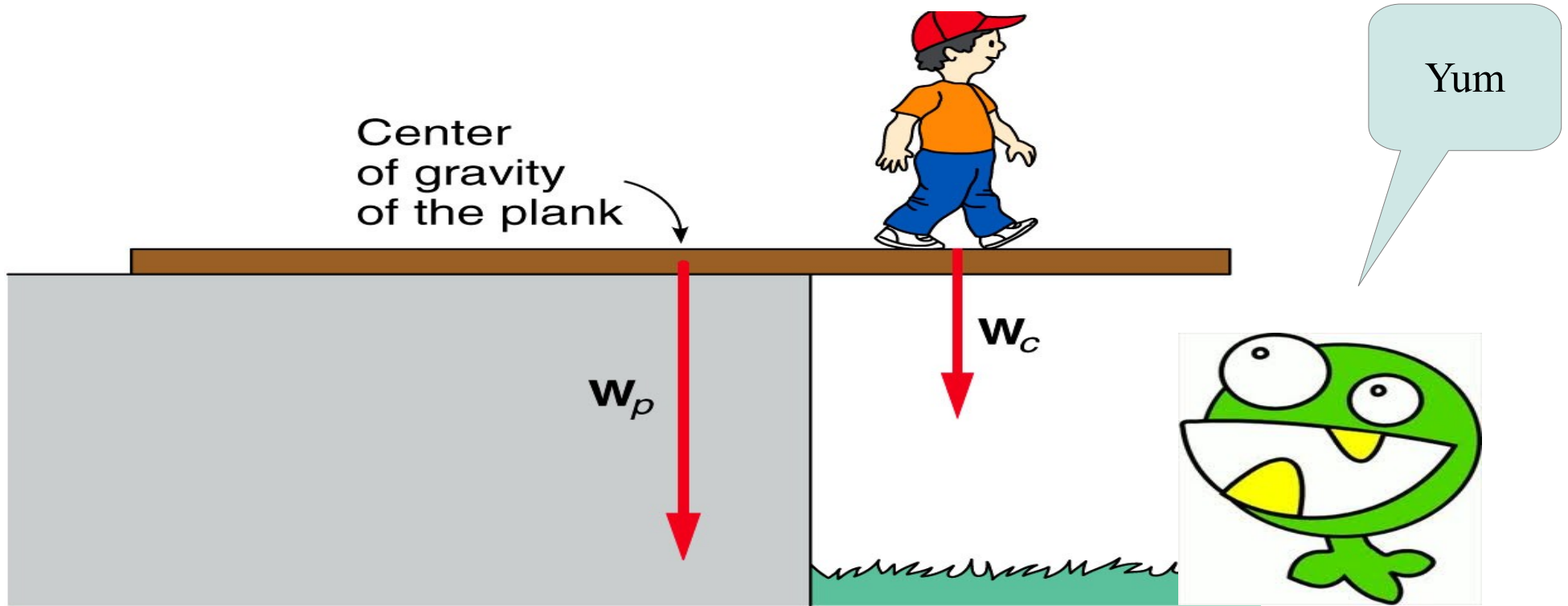
If you use this notation, you don't need to convert to meters or to newtons
As long as all the forces are expressed in the same unit and the
Distances are expressed in the same unit.

+ if counterclockwise motion and – if clockwise motion about the axis

II. $\sum (F_y) = 0$ up means $F_y > 0$ and down means $F_y < 0$

III. $\sum (F_x) = 0$ right means $F_x > 0$ and left means $F_x < 0$

OTHER example of equilibrium:

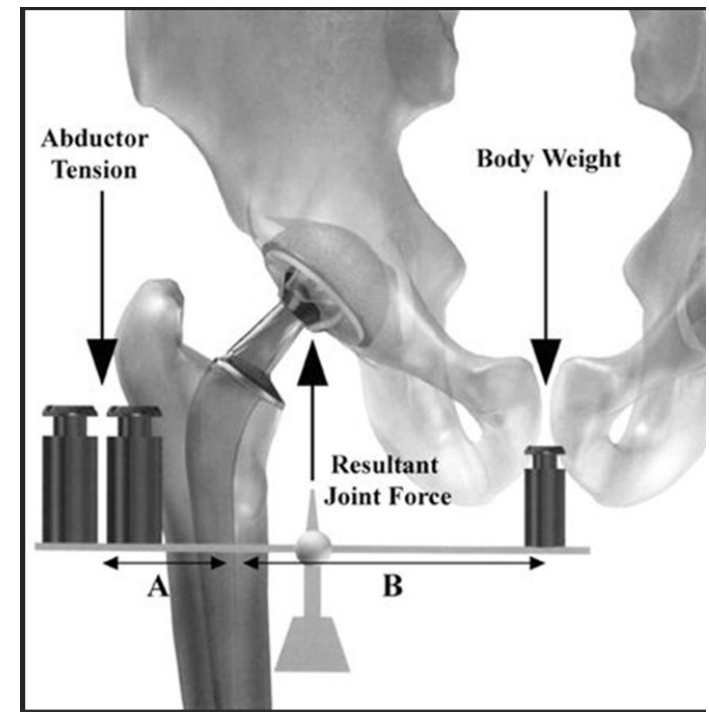
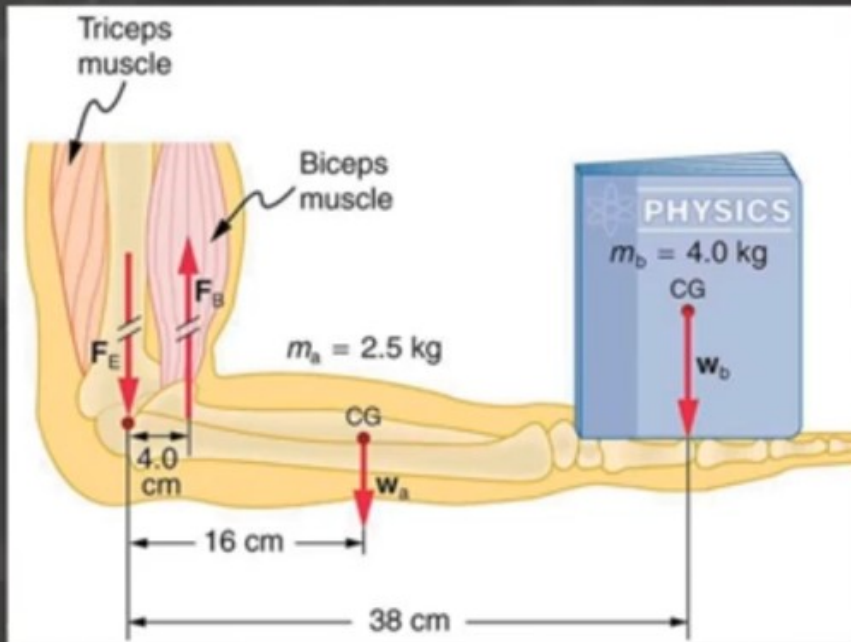


A 4m long plank with a weight of 80lb is placed on a dock with 1 m of its length extended Over a sea of deadly grass. A boy with a weight of 150lb is standing on the plank and moving out slowly from the Edge of the dock.

A) What is the torque exerted by the weight of the plank about the pivot point at the edge of the dock ? (treat all the weights as acting through the center of gravity of The plank).

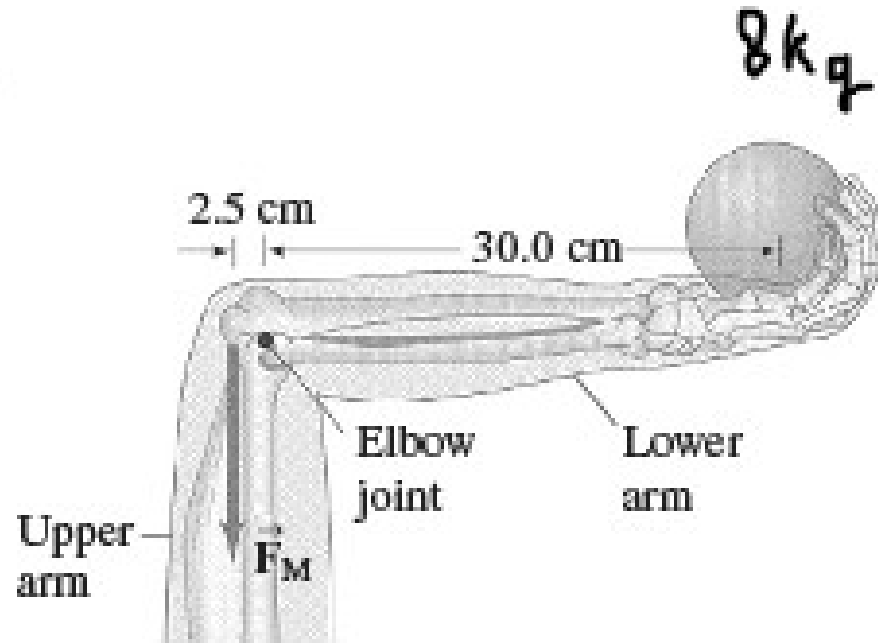
B) How far from the edge of the dock can the boy move until the plank is just on the verge Of tipping ? (and the boy on the verge to be eaten)

$\text{Torque} = \text{Force} * \text{Lever Arm}$



<https://www.barbellmedicine.com/blog/barbell-forces-levers-moments-and-torque/>

An athlete holds a 8-kg shot put in his hand with his lower arm horizontal, as shown in the figure. His lower arm has a mass of 3 kg and its center of gravity (or center of mass) is 12 cm from the elbow-joint pivot. How much force must the extensor muscle (which is M in the figure) in the upper arm exert on the lower arm?



Rule: **if distance effort-fulcrum > distance load-fulcrum → force amplifier**

Rule: **if distance load-fulcrum > distance effort-fulcrum → distance amplifier**

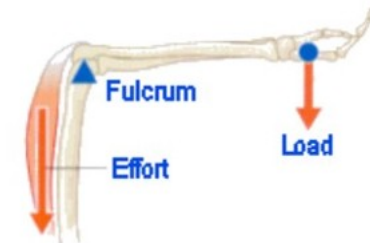
If same distance → not amplifier. Just changes the direction.

	Class 1 lever	Class 2 lever	Class 3 lever
Diagram			
Real world example			
Joint example			

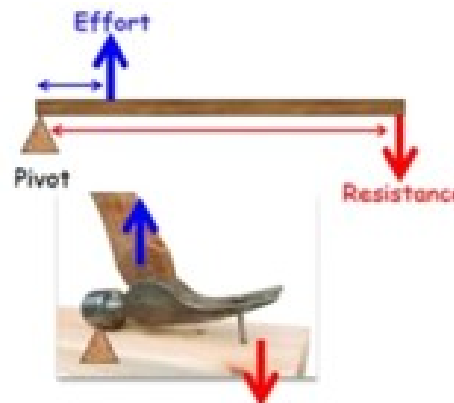
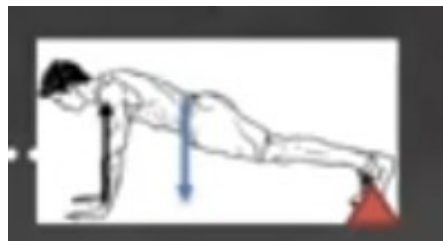
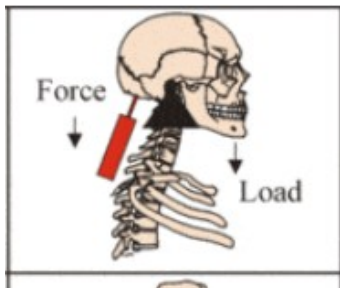
Type I: can amplify force or distance

Type II: amplify force only

Type III: amplify distance only

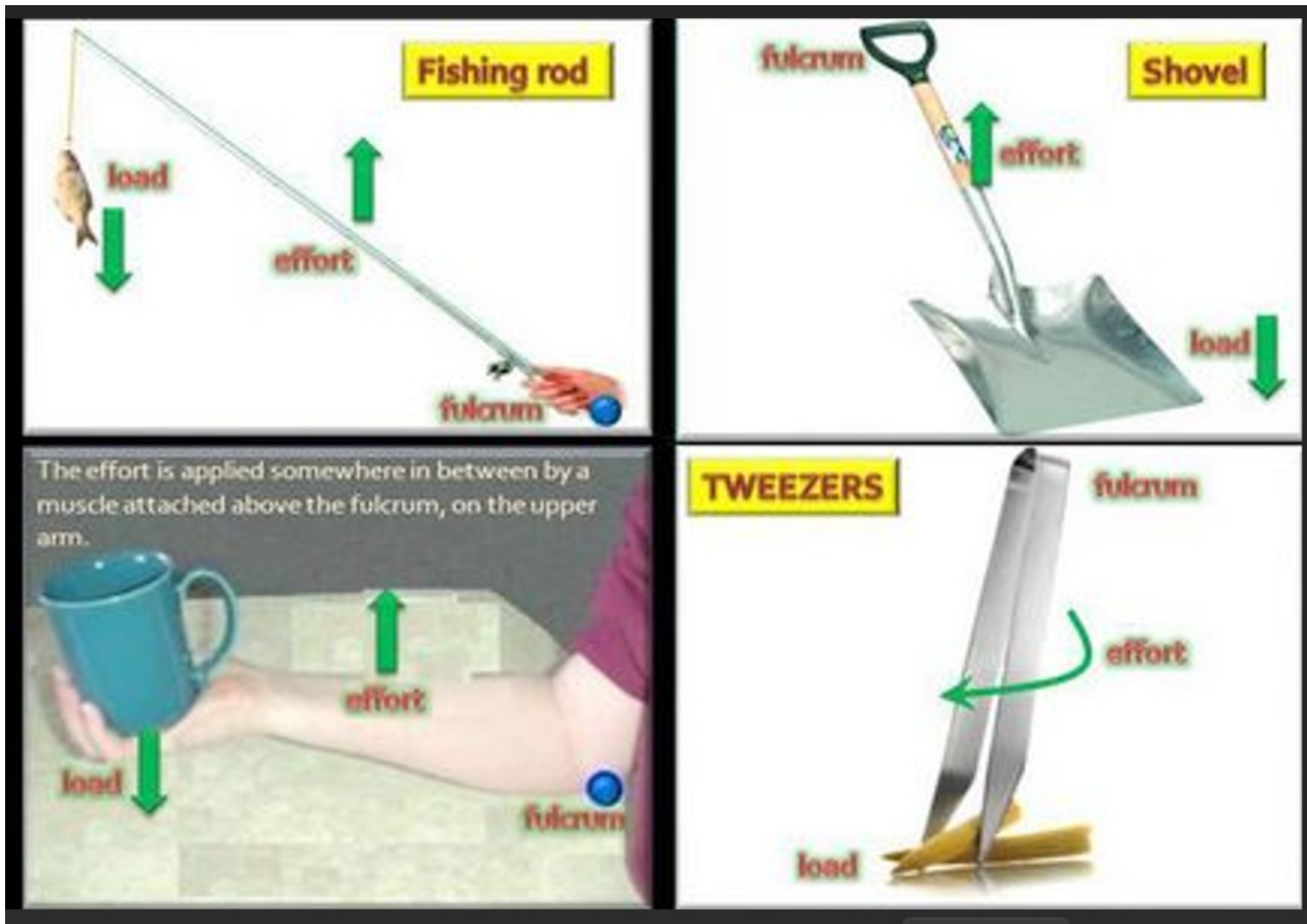


Class 1 Lever in the Body



This amplifies motion.
Biceps extension.
This is class 1 lever.

Type 3: amplify the distance



The Musculoskeletal System Includes...

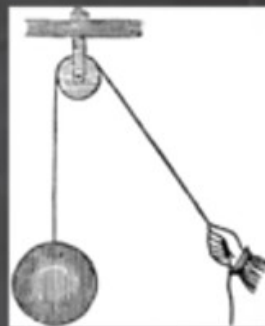
- Muscles – generate force
- Tendons – transfer force to bones
- Bones – move if enough force is transmitted
- Joints – allow bones to move

In Biology, these components form the **musculoskeletal system**.

In Physics, these components form a system of **simple machines**.

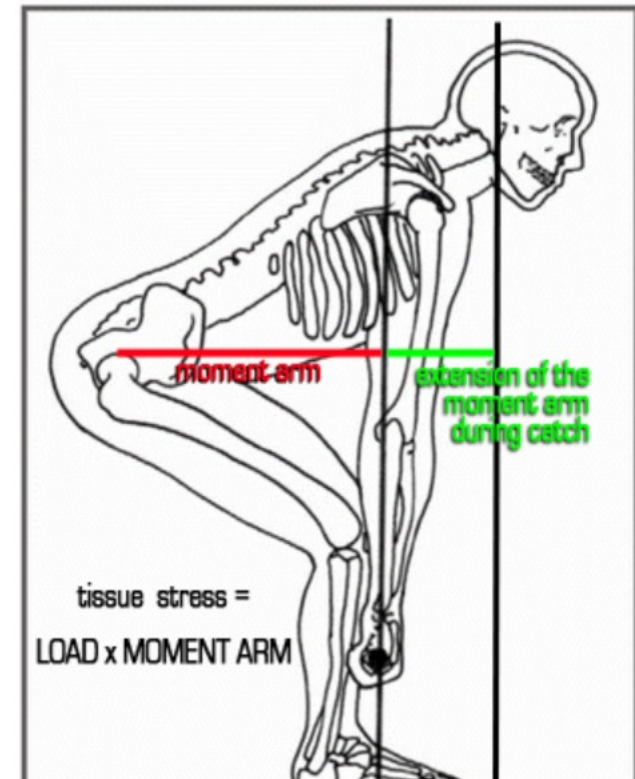
- The musculoskeletal system uses three types of simple machines

- Levers
- Wheel-axes
- Pulleys



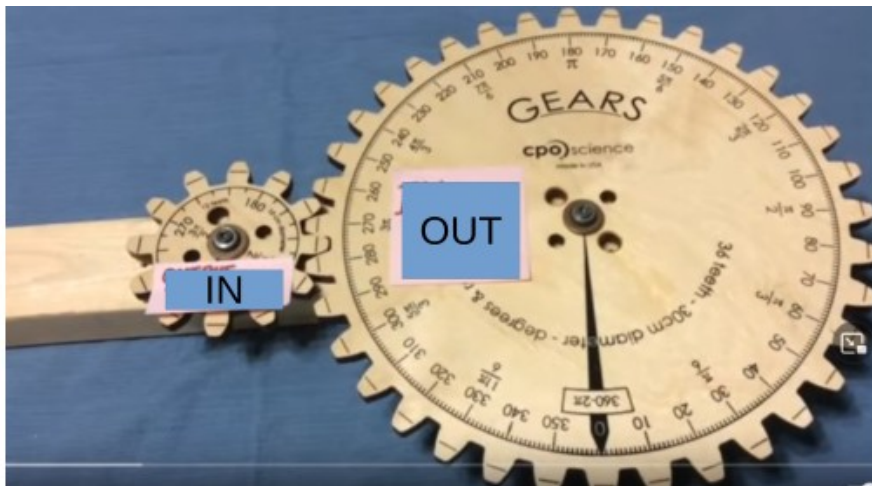
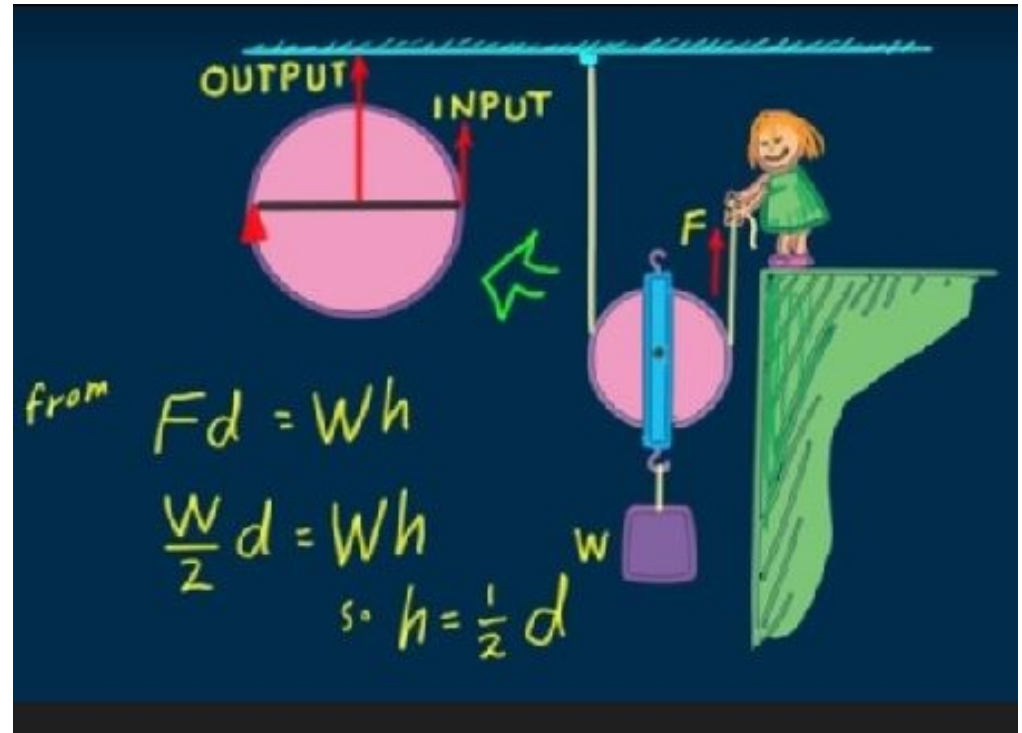
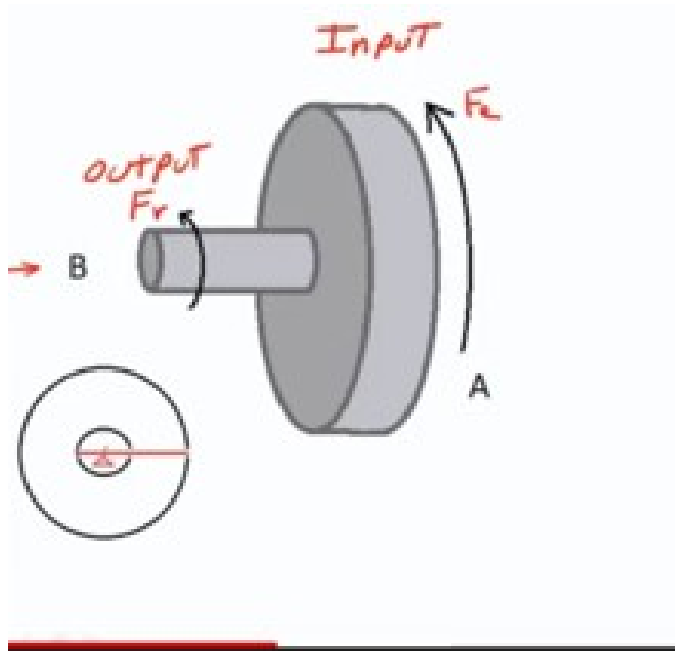
Biomechanics

<https://www.slideshare.net/slideshow/hip-biomechanics/28066355>



<https://www.slideserve.com/tuyen/biomechanics-of-musculoskeletal-system-pp-420-436-objectives>

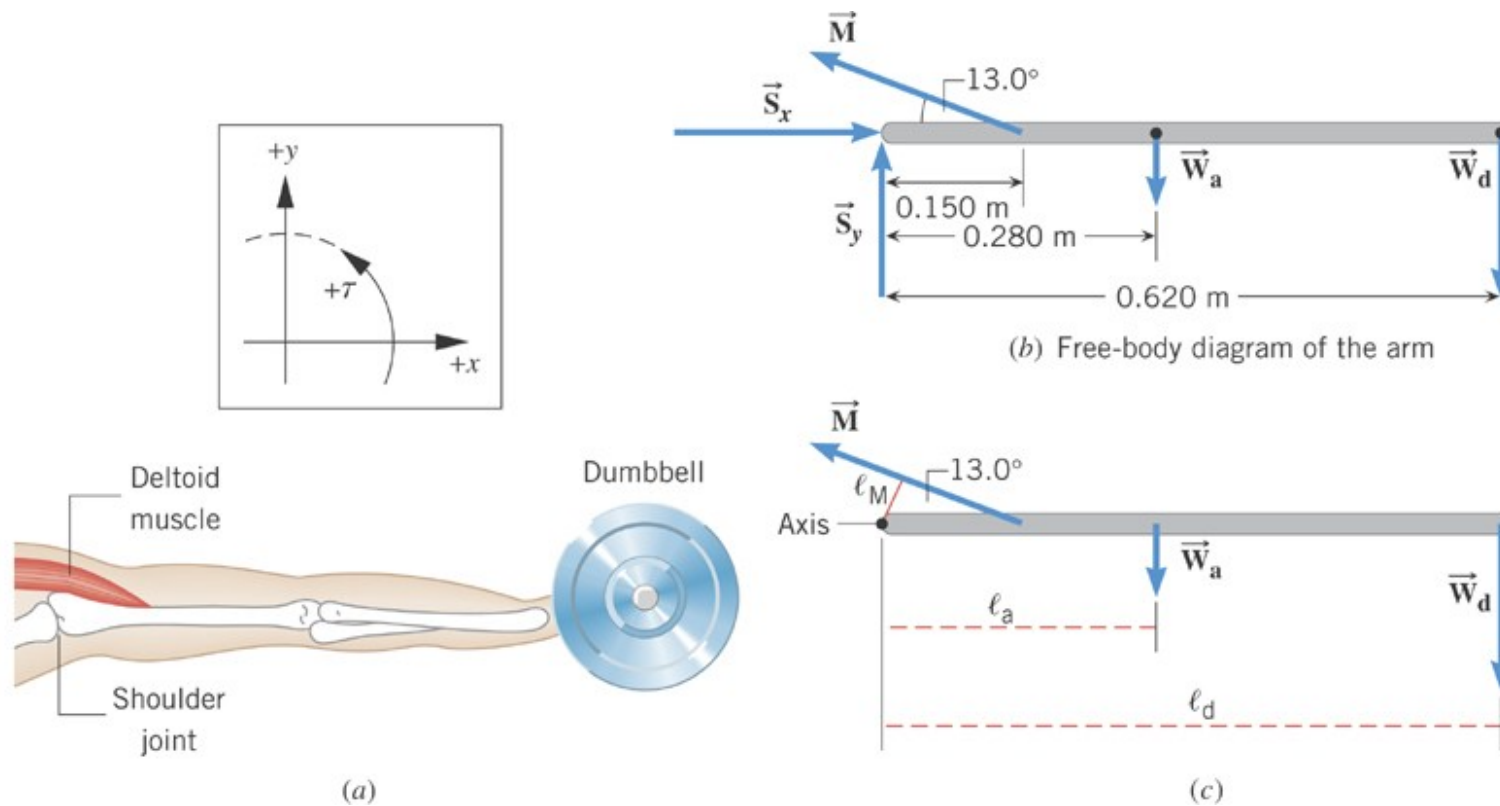
Screw driver



<https://bike.bikegremlin.com/104/bicycle-gear-ratios-speeds-gear-inches/>

Can amplify speed or force.

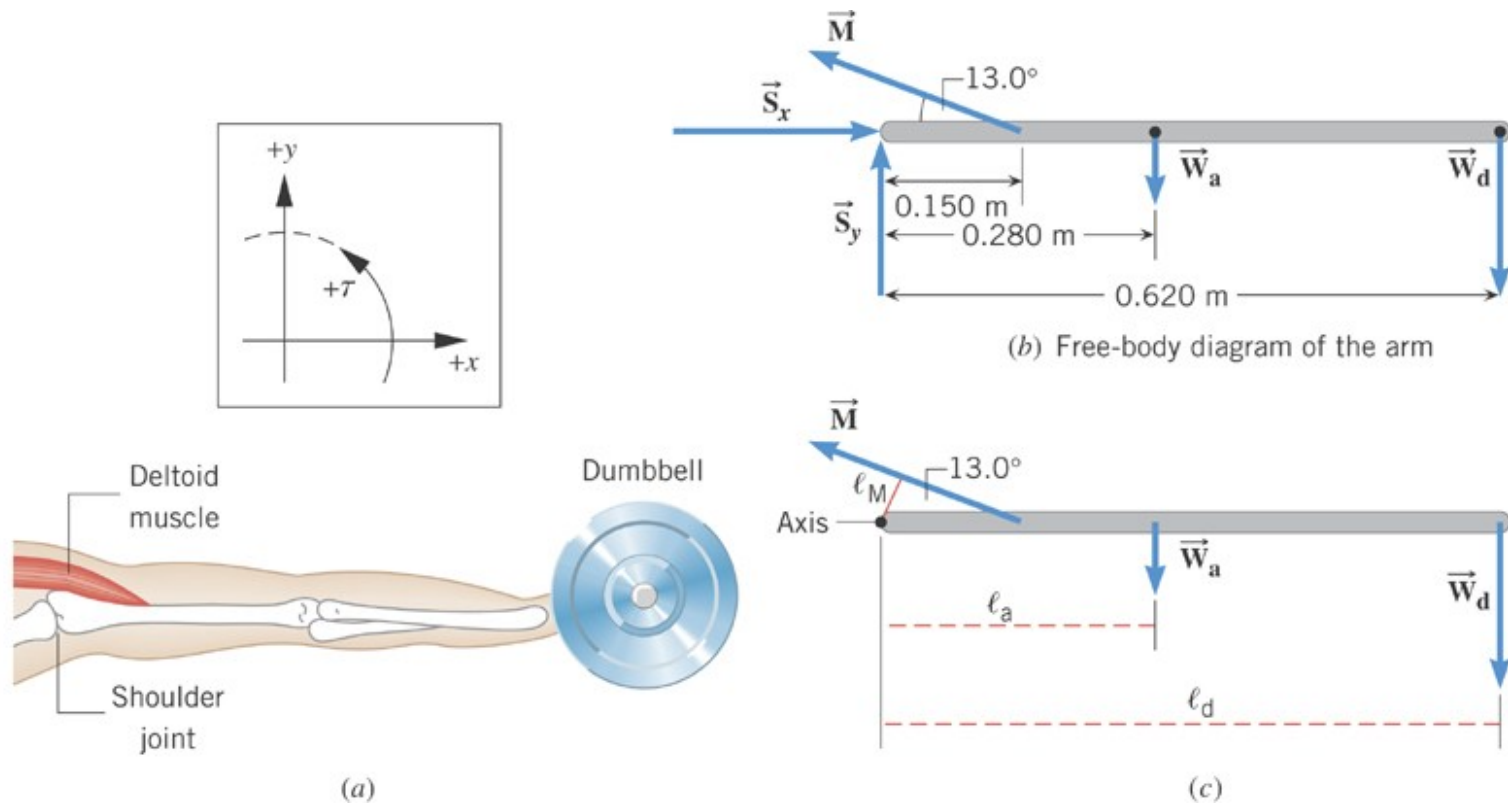
9.2 Rigid Objects in Equilibrium



Example 5 Bodybuilding

The arm is horizontal and weighs 31.0 N. The deltoid muscle can supply 1840 N of force. What is the weight of the heaviest dumbbell he can hold?

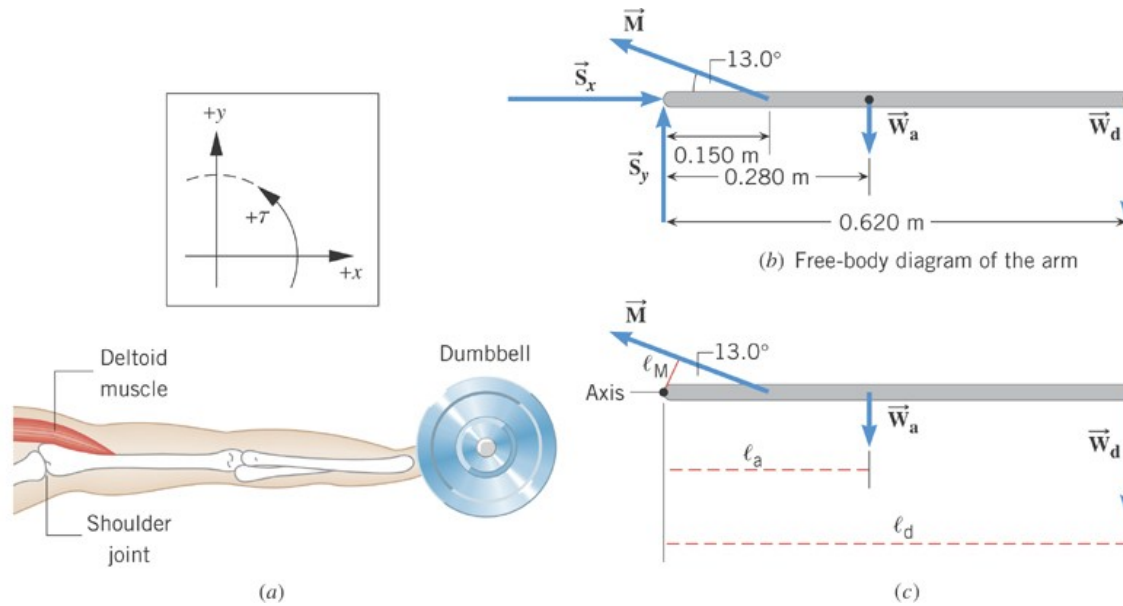
9.2 Rigid Objects in Equilibrium



$$\sum \tau = -W_a \ell_a - W_d \ell_d + M \ell_M = 0$$

$$\ell_M = (0.150\text{ m}) \sin 13.0^\circ$$

9.2 Rigid Objects in Equilibrium



$$W_d = \frac{-W_a \ell_a + M \ell_M}{\ell_d}$$

$$= \frac{-(31.0 \text{ N})(0.280 \text{ m}) + (1840 \text{ N})(0.150 \text{ m}) \sin 13.0^\circ}{0.620 \text{ m}} = 86.1 \text{ N}$$

9.2.3. Consider the three situations shown in the figure. Three forces act on the triangular object in different ways. Two of the forces have magnitude F and one of the forces has a magnitude $2F$. In which case(s), if any, will the object be in equilibrium? In each case, the forces may act at the center of gravity or at the center of a corner.

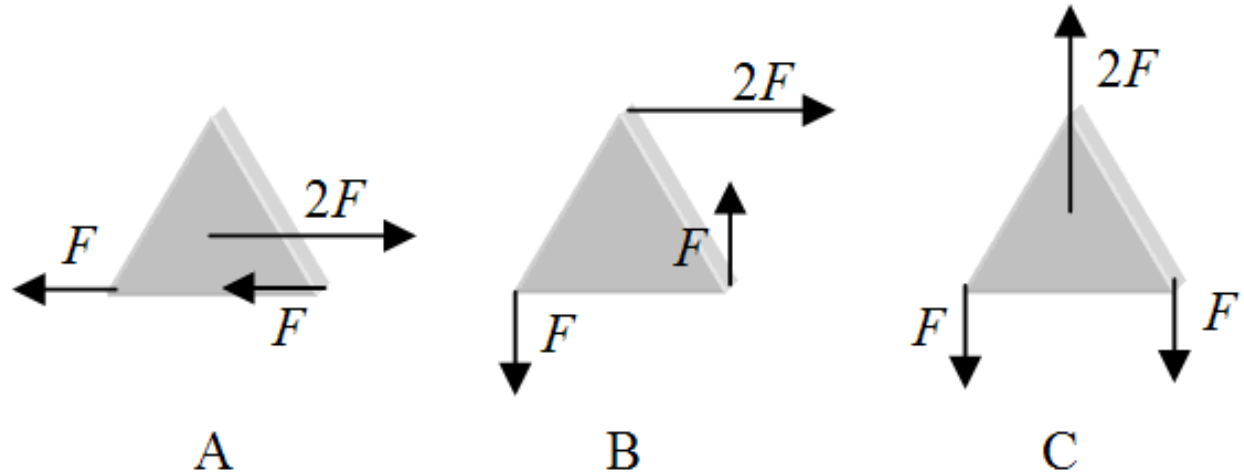
a) A only

b) B only

c) C only

d) A and C

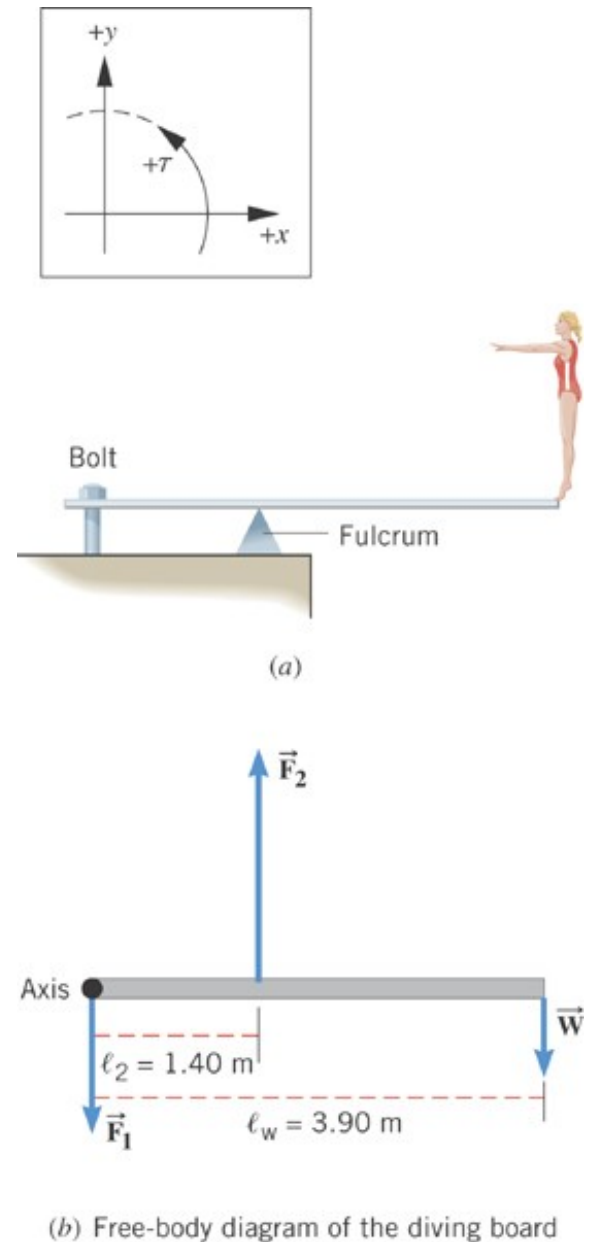
e) A and B



Example 3 A Diving Board

A woman whose weight is 530 N is poised at the right end of a diving board with length 3.90 m. The board has negligible weight and is supported by a fulcrum 1.40 m away from the left end.

Find the forces that the bolt and the fulcrum exert on the board.

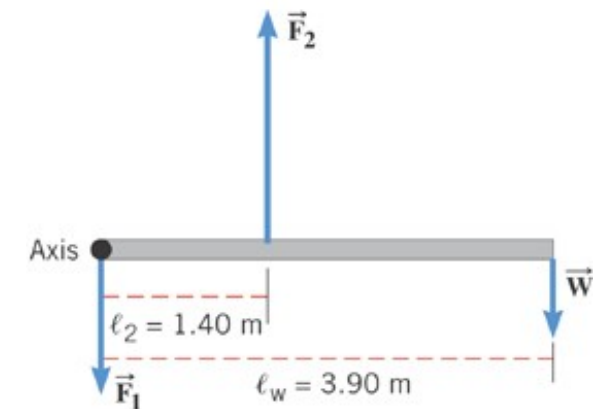
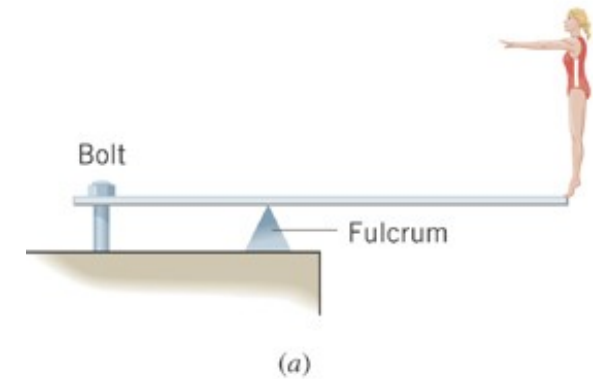
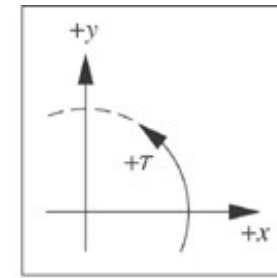


9.2 Rigid Objects in Equilibrium

$$\sum \tau = F_2 \ell_2 - W \ell_W = 0$$

$$F_2 = \frac{W \ell_W}{\ell_2}$$

$$F_2 = \frac{(530 \text{ N})(3.90 \text{ m})}{1.40 \text{ m}} = 1480 \text{ N}$$



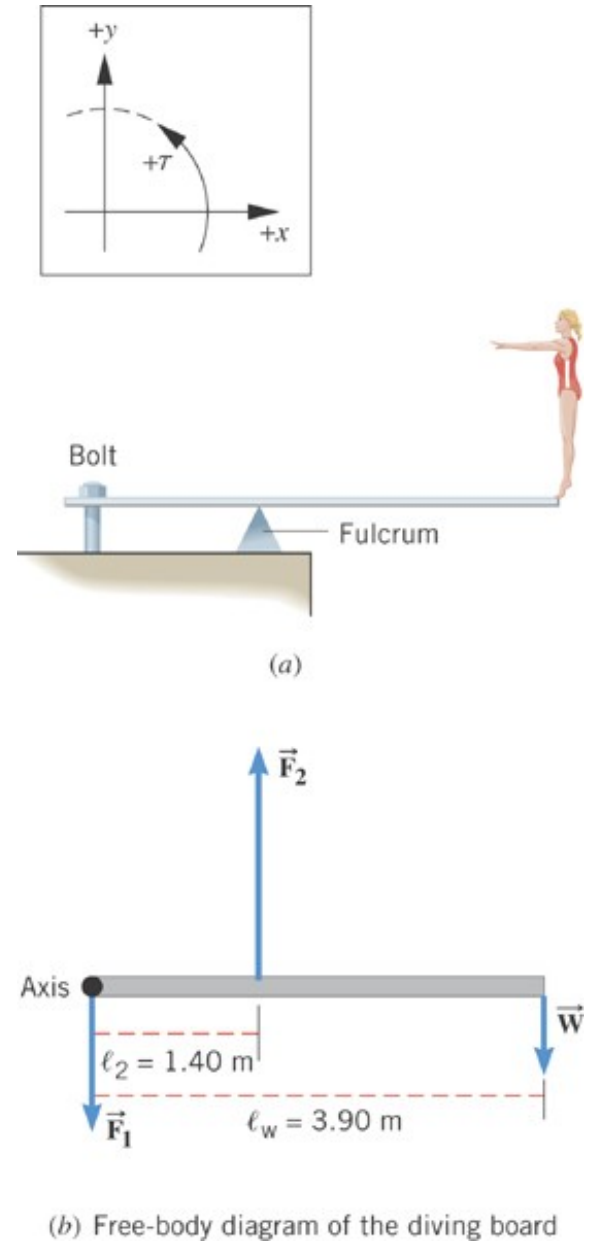
(b) Free-body diagram of the diving board

9.2 Rigid Objects in Equilibrium

$$\sum F_y = -F_1 + F_2 - W = 0$$

$$-F_1 + 1480 \text{ N} - 530 \text{ N} = 0$$

$$F_1 = 950 \text{ N}$$



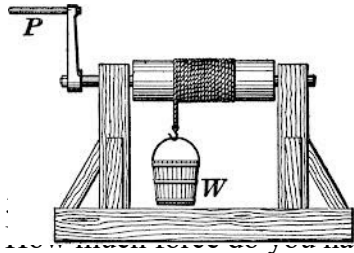
1) A weight of 30N is located at a distance of 10cm from the fulcrum of a simple balance beam. At what distance From the fulcrum should a weight of 20N be placed on the opposite side in order to balance the system ?

2) A weight of 5N is located at 10cm from the fulcrum of a simple balance beam. What weight should be placed at a point 4 cm from the fulcrum on the opposite side in order to balance the system.

3) A seesaw is 12 ft long and is pivoted at its center. George weighs 40 lb and is sitting at one end. His sister Alice, weighing 32 lb is 1 ft from him. Where does their 155 lb dad have to sit to balance the seesaw ?



4) The bucket (see figure) with the water in it has a mass of 6.0kg. The radius of the drum on which the rope is wound is 12cm. The length of the crank is 35 cm. How much force needed to lift the bucket ? (equilibrium between the 2 torques)



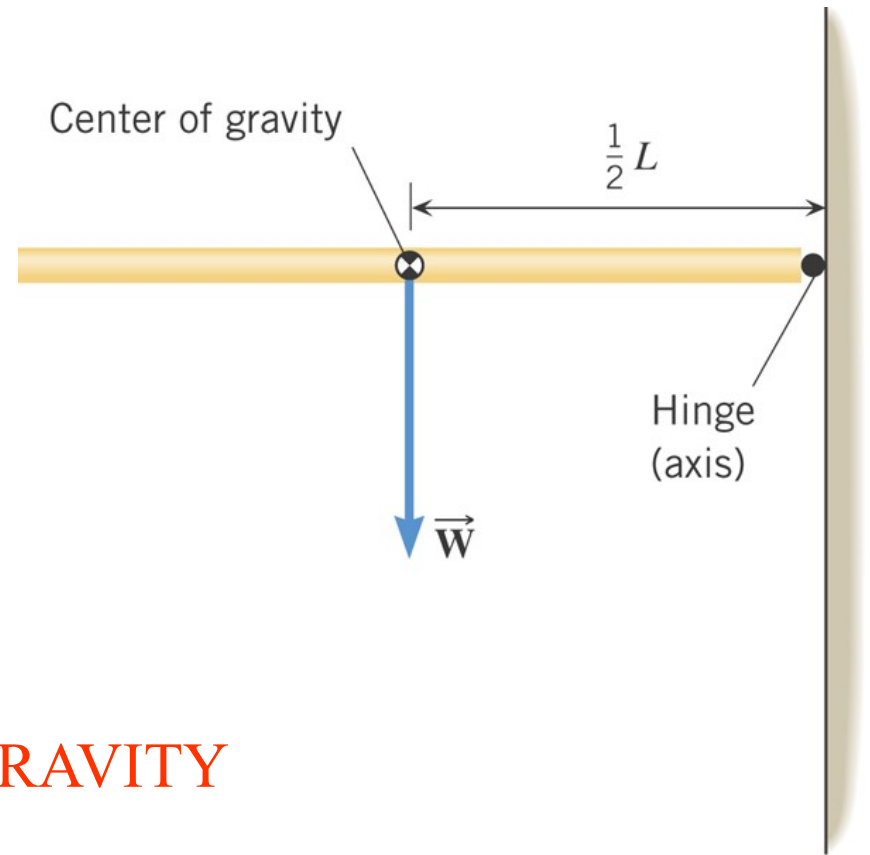
5) A plank 2.4 m long, pivoted at 0.2m from the end, to lift, at that end, a rock that weighs 750N. How much force do you have to exert ?

6) You are free to choose a fulcrum (pivot) where it is most convenient to do computation. Take the example of a plank with a pile of books (see figure). The fulcrum can be either of the chairs. The mass of the books is 35kg and the plank is 3.7m long, and the books are centered 1.2 m from the left chair. What is the elastic recoil (that is the normal force). You find the solution in 2 steps. First take the fulcrum at the right chair and Equilibrium equation. Then take the fulcrum at the left chair.



7) A crowbar 2.0m long is used to pry a rock out of the ground. By pivoting the crowbar 0.5m from the rock and pushing down on the other end. The end is pushed with a force of 250N. A) find the torque applied to the crow bar by the person doing the job. B) The force being applied to the rock.

9.3 Center of Gravity



DEFINITION OF CENTER OF GRAVITY

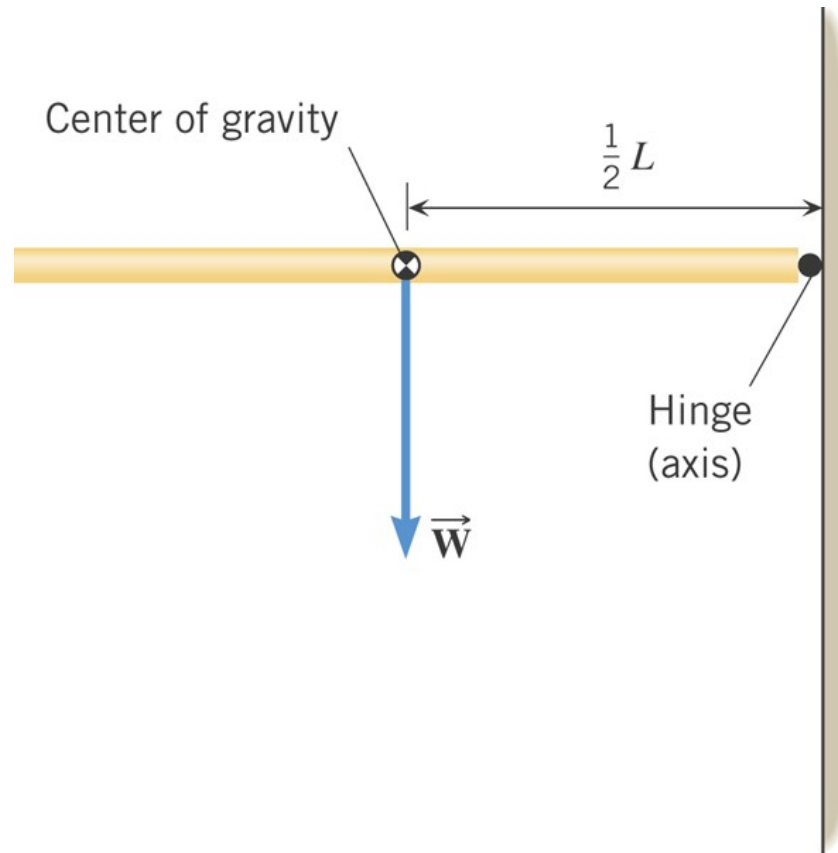
The center of gravity of a rigid body is the point at which its weight can be considered to act when the torque due to the weight is being calculated.

The Center of Mass

- There is a special point in a system or object, called the *center of mass*, that moves as if all of the mass of the system is concentrated at that point
- The CM of an object or a system is the point, where the object or the system can be **balanced** in the uniform gravitational field

9.3 Center of Gravity

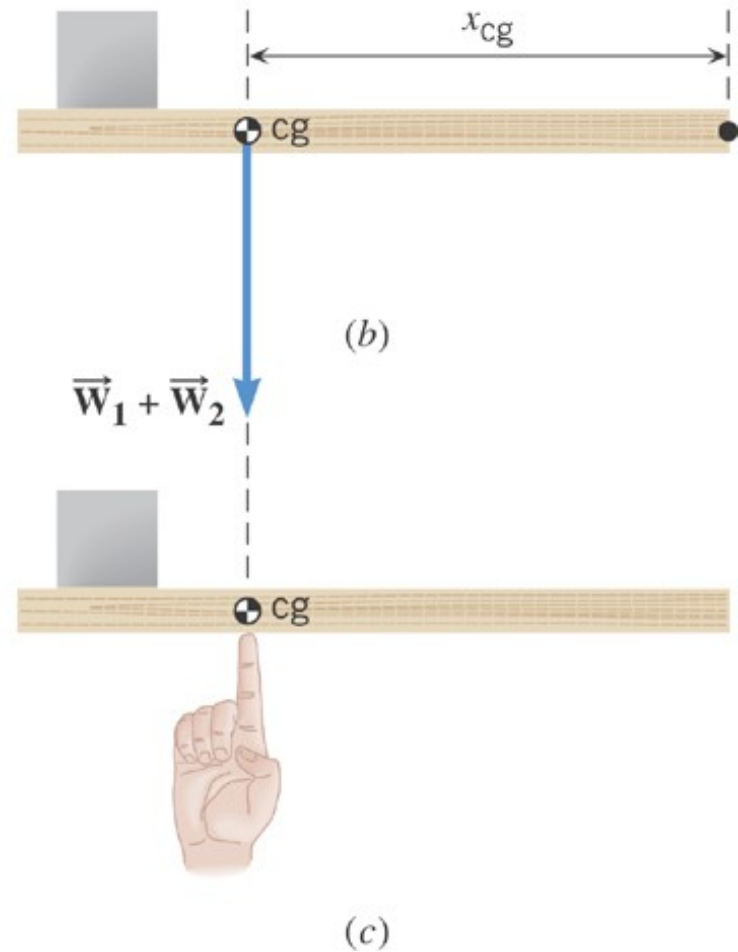
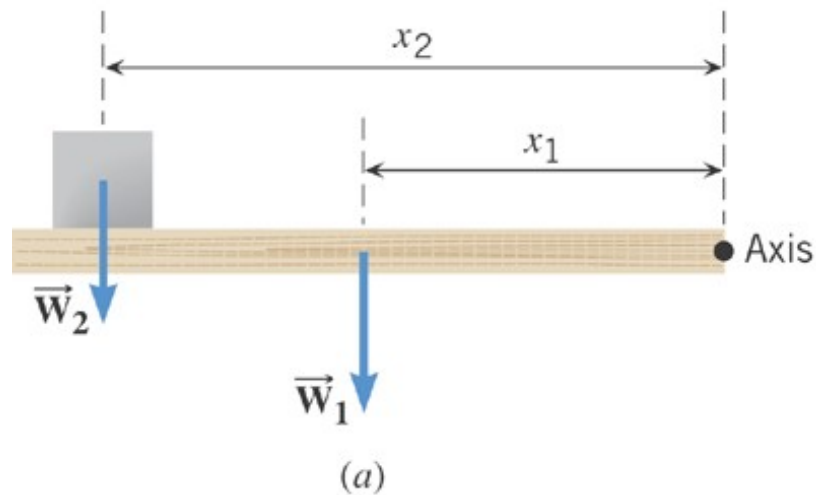
When an object has a symmetrical shape and its weight is distributed uniformly, the center of gravity lies at its geometrical center.



9.3 Center of Gravity

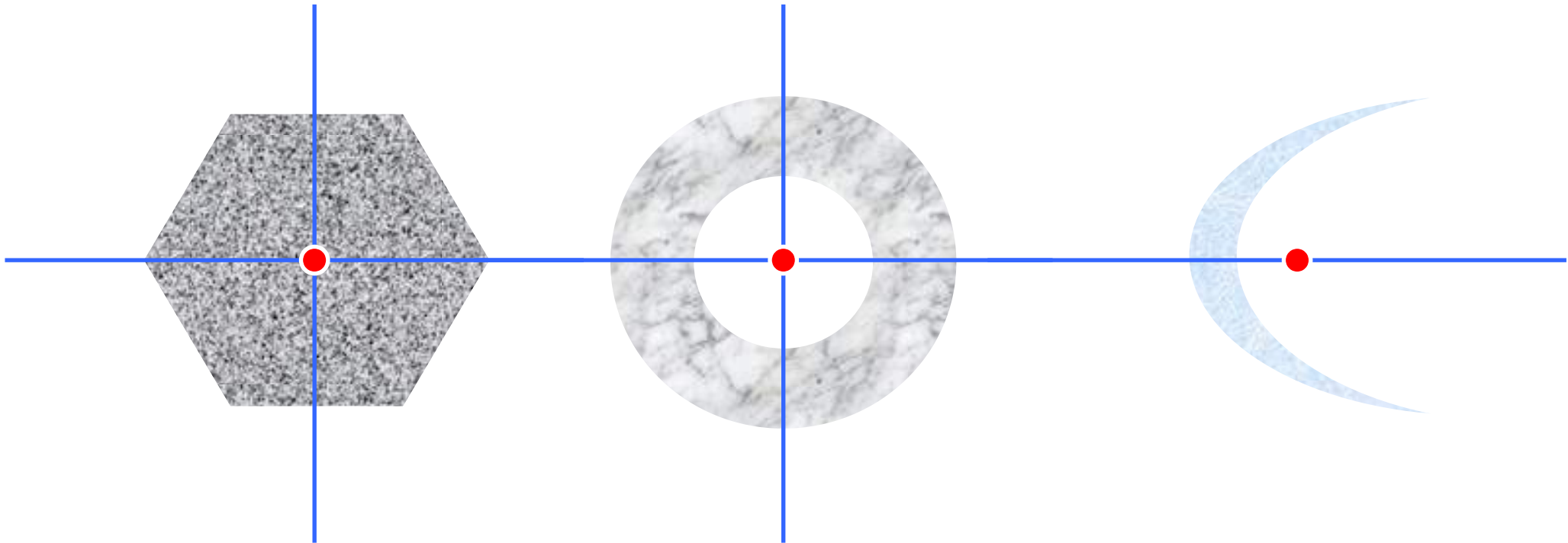
$$x_{cg} = \frac{W_1 x_1 + W_2 x_2 + \dots}{W_1 + W_2 + \dots}$$

Textbook 10.11

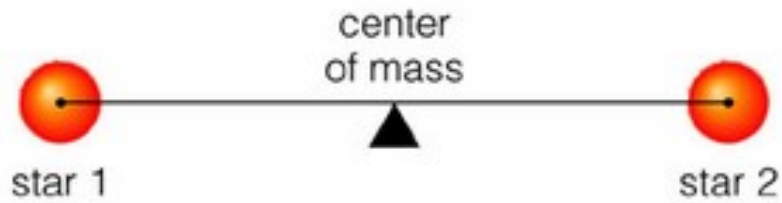


The Center of Mass

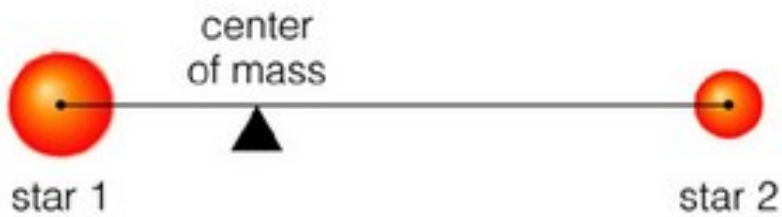
- The center of mass of any symmetric object lies on an axis of symmetry and on any plane of symmetry
 - If the object has uniform density
- The CM may reside inside the body, or outside the body



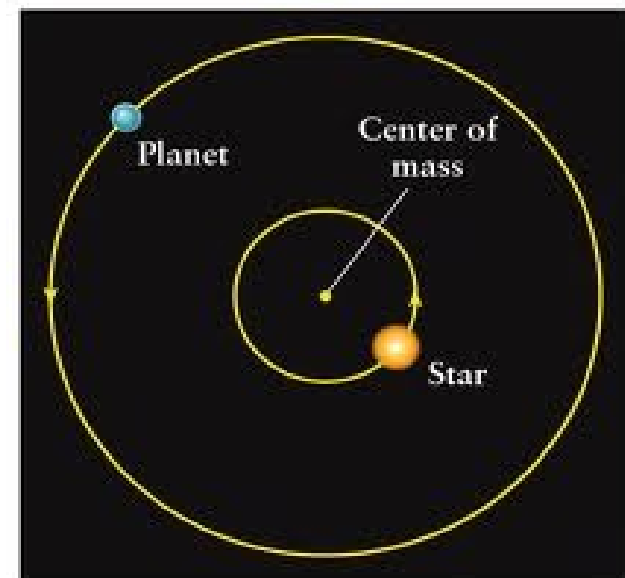
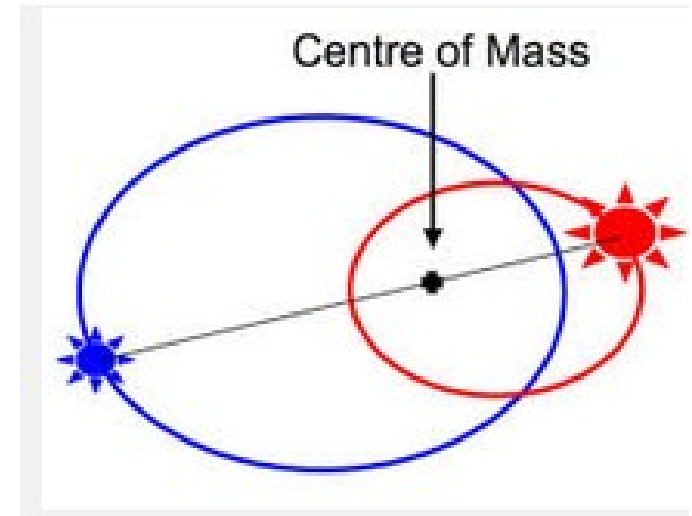
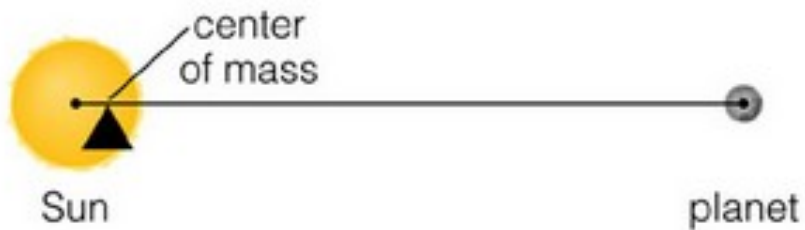
Two Stars of Equal Mass



Star 1 Is More Massive Than Star 2



Sun Is Much More Massive Than Planet



Movies in folder and MIT movie.



<https://www.zachbeach.com/the-physics-of-crow-pose/>

All objects behave as though their mass is concentrated at their center of gravity.

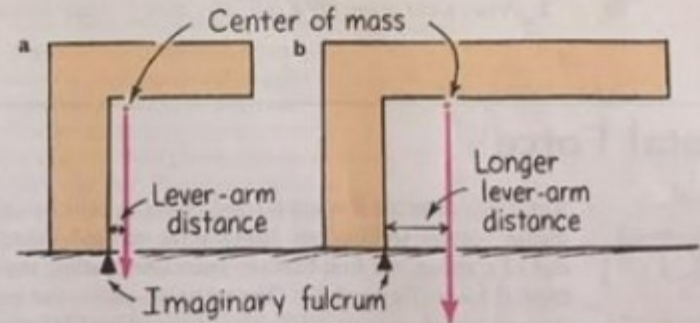
an object is stable as long as its center of gravity is in-line with the BASE of the object.

FIGURE 8.33
Where is Alexei's center of gravity relative to his hands?

Physics Place
Toppling



FIGURE 8.34
The greater torque acts on the figure in (b) for two reasons. What are they?



October 24, 2021

y University

THE EDGE IN KNOWLEDGE

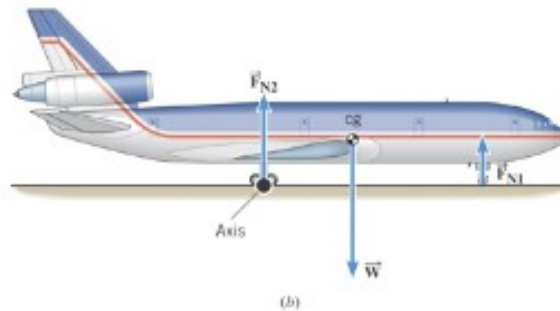


***Conceptual Example 7* Overloading a Cargo Plane**

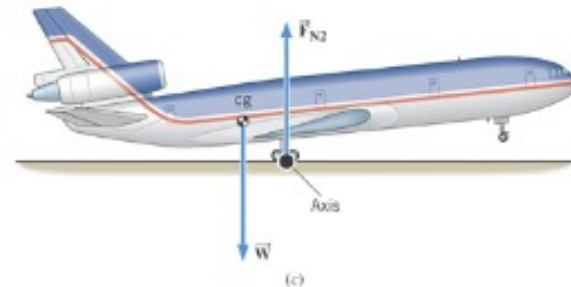
This accident occurred because the plane was overloaded toward the rear. How did a shift in the center of gravity of the plane cause the accident?



(a)



(b)



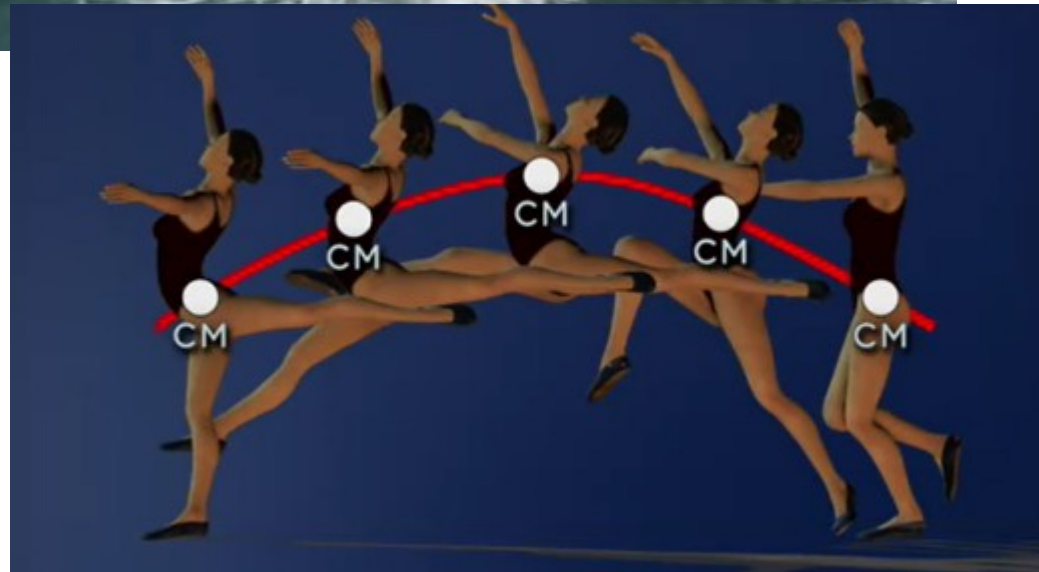
(c)

You can manipulate the CM.

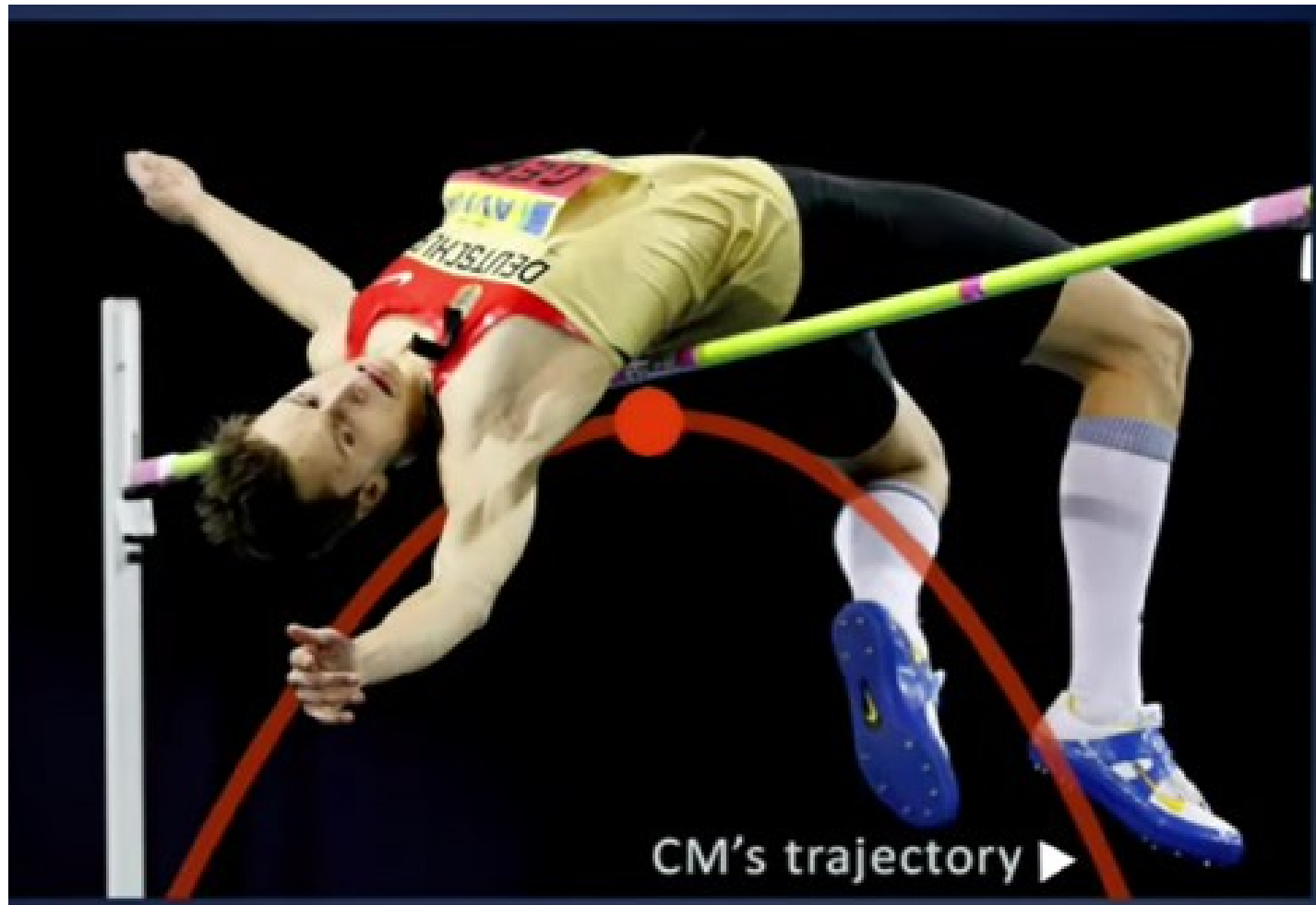
The CM responds only to the exterior force $\rightarrow$ gravity . It follow a Projectile motion (parabola). But by crunching, your CM moves to Upper body. So the body moves so the CM stays on the parabola. It looks like you are floating vs going up and down.



It looks like the
Head stays on
A horizontal
Because the CM
Is higher
(legs up)



The CM does not go higher compared to when the athlete jumps differently
But your body goes higher because you arch. The CM only obeys gravity.
Free-fall motion.





██████████ YOU GRAVITY

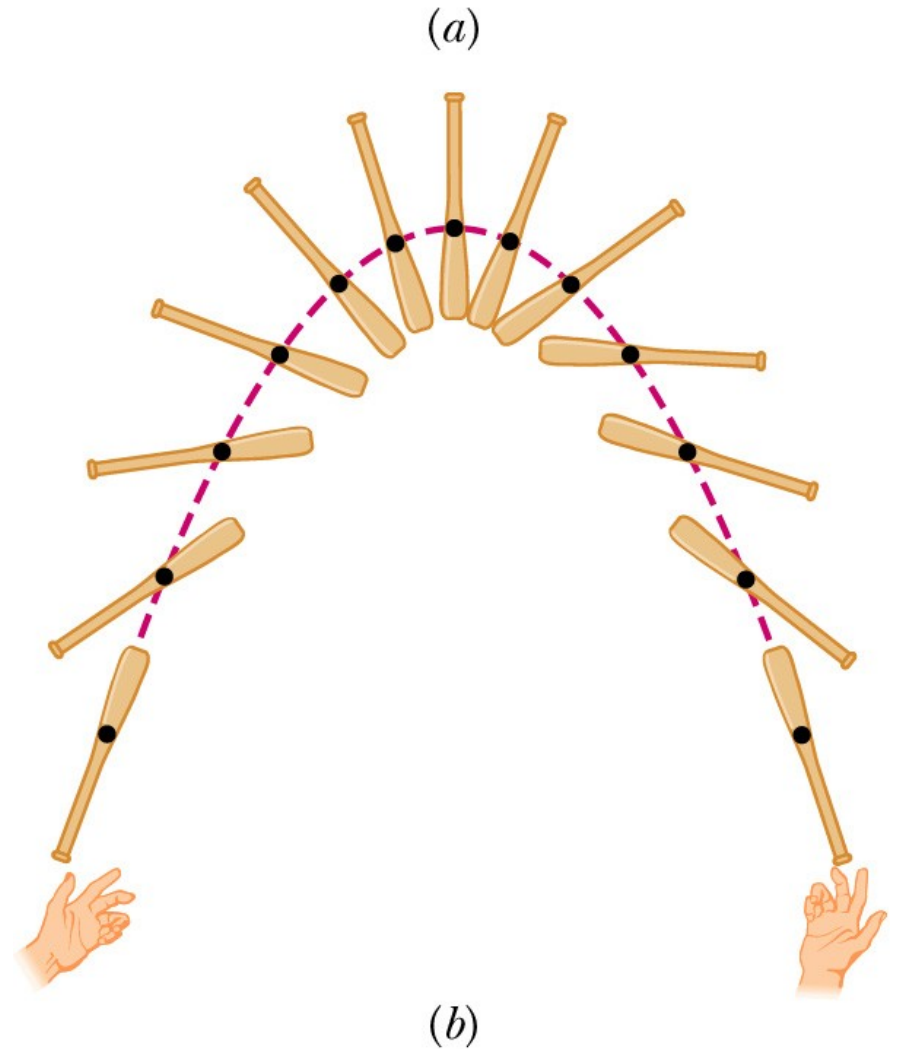
I'M MICHAEL JACKSON

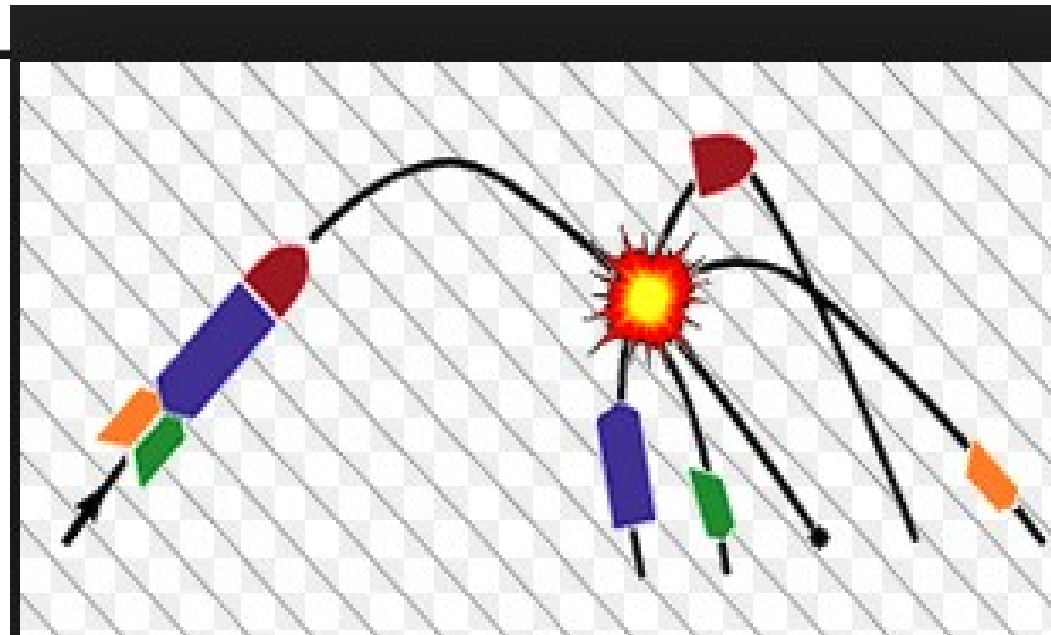
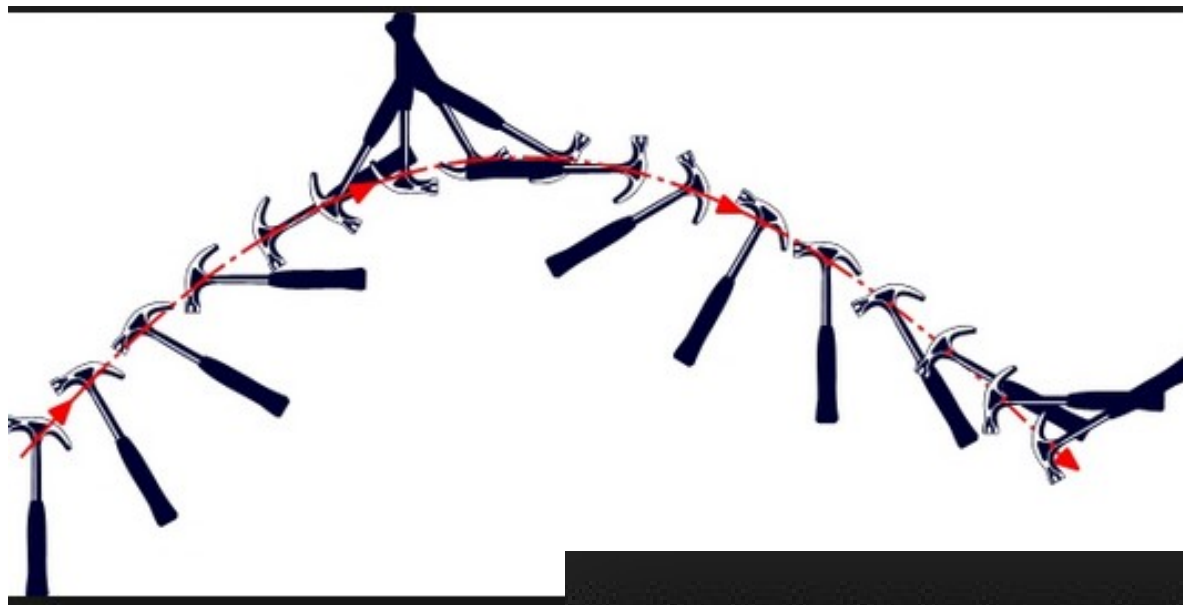


The Center of Mass

- How should we define the position of the moving body ?
- What is y for $U_g = mgy$?
- Take the average position of mass. Call “Center of Mass” (COM or CM)

Video MIT





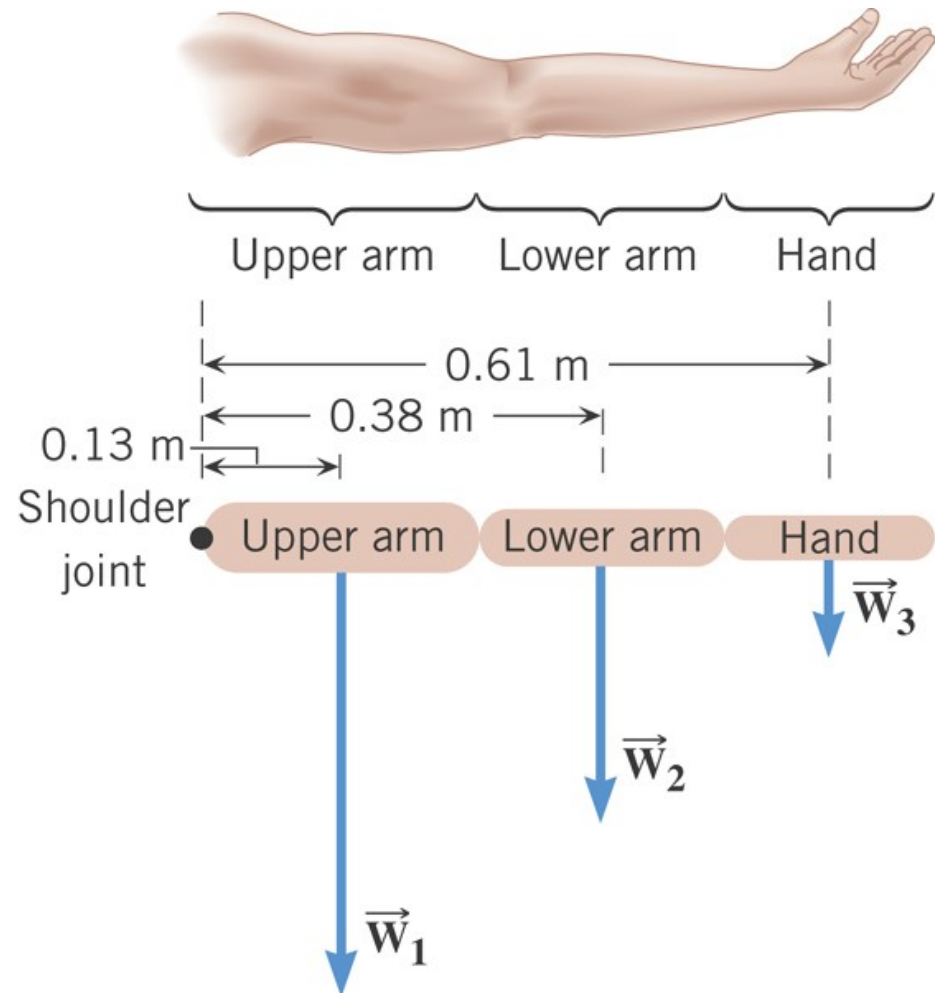
Center of Mass for many particles in 3D?



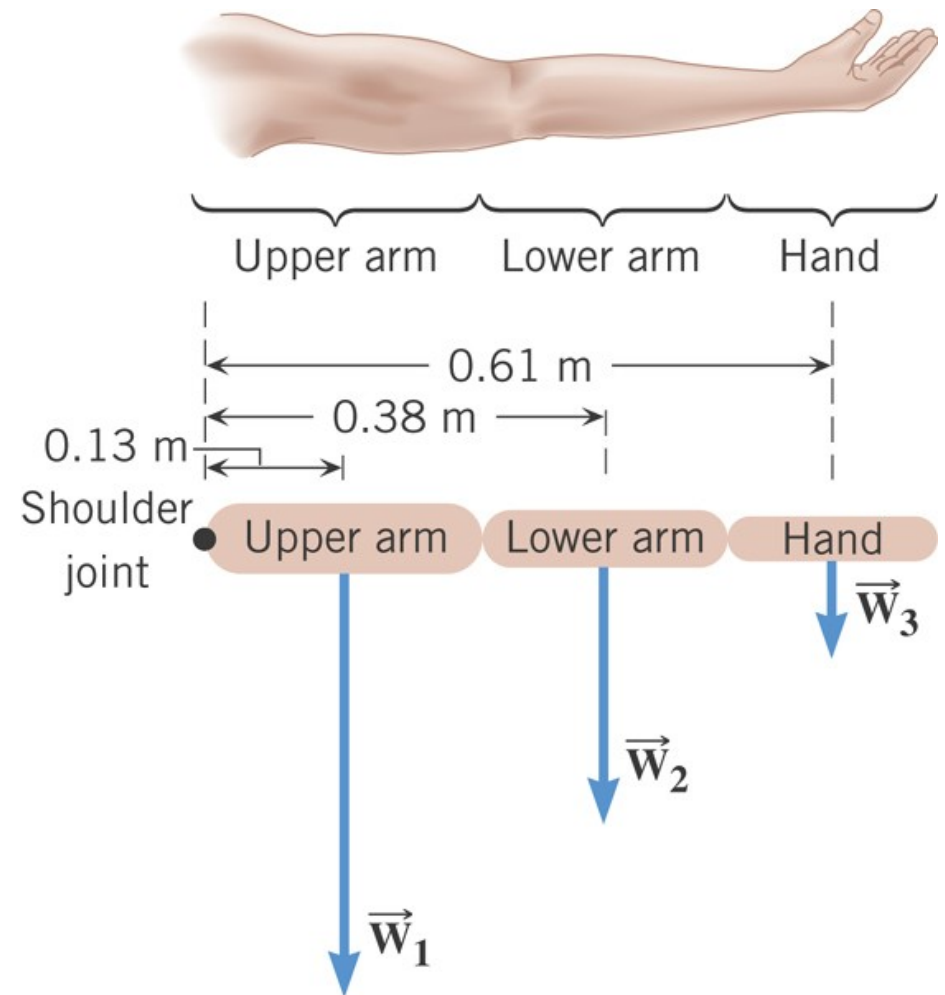
Example 6 The Center of Gravity of an Arm

The horizontal arm is composed of three parts: the upper arm (17 N), the lower arm (11 N), and the hand (4.2 N).

Find the center of gravity of the arm relative to the shoulder joint.



9.3 Center of Gravity



$$x_{cg} = \frac{W_1 x_1 + W_2 x_2 + \dots}{W_1 + W_2 + \dots}$$

$$x_{cg} = \frac{(17 \text{ N})(0.13 \text{ m}) + (11 \text{ N})(0.38 \text{ m}) + (4.2 \text{ N})(0.61 \text{ m})}{17 \text{ N} + 11 \text{ N} + 4.2 \text{ N}} = 0.28 \text{ m}$$

REVIEW spinning

5) suppose that a merry-go-round is rotating at the rate of 10 rev/min

A) Express this rotational velocity in rev/s (rev means revolution = 1 tour)

B) Express this rotational velocity in rad/s (1 revolution = 2π rad)

6) The popular music record used to have a rate of 45RPM .

A) express this velocity in rev/s

B) Through how many revolutions does the record turn in a time of 5s?

(its a rate. Use proportion)

7) Suppose a disk rotates through 3 revolutions in 4 seconds

A) What is its displacement in radians in this time ? ((1 revolution = 2π rad)

B) What is its average rotational velocity in rad/s

8) A bicycle wheel is rotation ally accelerated at the constant rate of 1.2 rev/s/s

A) If it starts from rest, what is its rotational velocity after 4s

B) Through how many revolutions does it turn in this time ?

(like for kinematics : # of revolution = $\frac{1}{2}$ (acceleration in rev/s/s) x (time squared) + (initial rotational velocity) x (time)

9) The rotational velocity of a merry go round increases at a constant rate from 1.0 rad/s to 1.8 rad/s in a time of 4s.

What is the rotational acceleration of the merry go round ?